Capability or Optimizer? What Lets an LLM Proposer Leave Its Template Family on Circle Packing

Anonymous authors
Paper under double-blind review

Abstract

Zero-shot LLM proposers on circle packing concentrate on a grid-plus-filler family whose best value follows in closed form from N. We ask what lets a proposer leave it. A fixed SLSQP program with no model in the loop clears the family in 45 of 45 runs and beats the best model-written program at every cell: the optimizer does the clearing. Across two proposer tiers and three output channels, the tiers differ in whether their programs drive it there. A Sonnet-tier program with `numpy` and `scipy` clears in 23 of 25 valid programs, or 23 of 41 once every invocation is charged; without the libraries, on the same path and budget, it clears none; a weak-tier program with them clears none of its 90 invocations, and 4 of 90 when its unreadable output is re-read leniently. Handed the better construction, the weak tier chooses it but builds it in 6 of 14 valid outputs: execution, not preference, is the bottleneck. The zero-shot characterization survives neither conditioning, nor five loop generations, nor a change of serving path, where the reasoning budget restores it. Twenty-six registered arms are reported, whichever way each went.

1 Introduction

Ask a language model to place 21 circles in a unit square to maximize the sum of their radii, and it will not search. It reaches for the nearest square grid, five by five at radius $1/(2k)$, and truncates it to 21 circles, leaving four cells empty. A better construction is one parameter away: a four-by-four grid with interior-vertex fillers of radius $(\sqrt{2}-1)/(2k)$, one in each gap between four grid circles. The grid-plus-filler family has a closed-form best value at every N; this paper asks what it takes for a proposer to leave it.

Circle packing is the showcase benchmark of LLM-driven discovery, which FunSearch (Romera-Paredes et al., 2024) opened. AlphaEvolve (Novikov et al., 2025) reported 2.63586276 at $n = 26$. Later systems cluster: ShinkaEvolve (Lange et al., 2025) 2.6359831, or 2.6359777 strict; HELIX 2.63598308 (Su et al., 2026); GigaEvo 2.63598 (Khrulkov et al., 2025); AdaEvolve 2.636 (Cemri et al., 2026); and ThetaEvolve (Wang et al., 2025) claims new best-known bounds. Every one pairs a capable model with an executed optimizer; two critiques from outside ask what each contributes: Gideoni et al. (2026) show simple baselines recover much of the reported advantage, and Berthold et al. (2026) show classical solvers do too. We ask the same question from inside, varying the two factors separately.

The result is the optimizer; the table locates it. Two tiers of one vendor, weak and Sonnet, cross three channels: direct coordinate emission, a program allowed only `math`, and a program allowed `numpy` and `scipy`. Table 1 scores every cell under one clearance rule. Every prediction whose verdict this section reports was registered before the arm's first scored run, with the arm-S exception Table 5 records; readings marked post hoc were not. Commits, amendments and outcomes are in the deviation table, Table 5. One cell clears at rate. A Sonnet-tier program with the libraries clears the family in 23 of 25 valid programs, at 3 of 3 cells (§3.5). Its median margin is 3.6% and its best per cell 2.5%, 3.6% and 4.2%. A weak-tier program with the same libraries clears 0 of 19 valid over 90 invocations, and its 95% Wilson upper bound of 16.8% sits under the registered 20% bar. A post hoc lenient re-read of its unreadable outputs recovers 4 clearing programs, 4 of 41 valid and 4 of 90 invocations. A Sonnet-tier program without the libraries clears 0 of 37 on the agent runtime and 0 of 35 on the library arm's (arm CL) own path and budget. The library's credit is that same-path gap,

Tier	Channel	Instrument	Cells	Clear	Note
weak	direct emission	agent runtime	13, 21, 31, 43	0 of 57	trap cells; at the extend cell $N = 20$ (arm M) 2 of 15 clear, both out-of-family
weak	direct emission	pinned API	7 square cells	0 valid of 105	unscoreable (arm P)
Sonnet	direct emission	agent runtime	13, 21, 31	1 of 30	one out-of-family packing at $N = 31$; 27 of 30 valid at 10^{-9}, the clearance among them (arm S)
weak	math-only program	agent runtime	13, 21, 31	0 of 78	reaches the argmax exactly 2 times, never passes it (CC, CC2)
weak	math-only program	pinned API	13, 21, 31	0 of 12	two cells under the five-valid floor; 0 of 6 at the scoreable cell (arm P code cells)
Sonnet	math-only program	agent runtime	13, 21, 31	0 of 37	annealing and multi-restart search, still 0 (CCS)
Sonnet	math-only program	pinned API	13, 21, 31	0 of 35	the library row's path and 120 s budget without the libraries; best per cell 1.625, 2.1, 2.673, none at the argmax (CCP)
weak	numpy/scipy program	pinned API	13, 21, 31	0 of 11; 0 of 19 pooled	arm CL, then pooled with its registered top-up CL-W (90 invocations; the $N = 21$ cell stays under the five-valid floor): Wilson upper bound 16.8%, under the 20% bar; a post-hoc reparse of the `numpy`-scalar stdouts reads 4 of 41 (CL, CL-W)
Sonnet	numpy/scipy program	pinned API	13, 21, 31	**23 of 25**	3 of 3 cells; excess over the argmax 0.67–4.24%, median 3.62% (CL)
none	numpy/scipy program, fixed	local, 120 s	13, 21, 31	**45 of 45**	a fixed random-restart SLSQP reference program, no model, through the library row's pipeline; 3 of 3 cells at the 20% bar; best per cell exceeds the Sonnet best at 3 of 3 cells (arm B)

Table 1: **Who clears the family argmax.** Each row is one arm or one registered pair of arms, scored under §2.4's clearance rule: valid at 10^{-9} and above the family argmax by more than 10^{-6}, the argmax recomputed in closed form. "Clear" counts valid outputs. Instrument names the serving path, which arm P shows is not interchangeable: the same weak-tier prompt yields 78% valid packings through the agent runtime and 0 of 105 through the pinned API. Rows are not pooled across instruments.

0 of 35 against 23 of 25 (the isolating arm, arm CCP). Two cells show low-rate escapes, both out-of-family constructions: one Sonnet direct emission in 30, and two weak-tier emissions in 15 at one extend cell, from a post hoc sweep. A fixed random-restart SLSQP program with no model in the loop clears 45 of 45 through the same pipeline and beats the best Sonnet program at every cell (the optimizer-alone baseline, arm B). The optimizer clears the family: the Sonnet tier's programs clear because they invoke it, and the weak tier's programs invoke it and do not clear.

Why: execution, not preference. Handed the better in-family construction, the weak tier builds it correctly in 6 of 14 valid outputs (the argmax-in-hand arm, arm CH). A post hoc attempt-level reading finds it reaching for that construction in 36 of 45 invocations. In the math-only channel the weak tier reaches the argmax construction exactly twice in 78 valid programs and never passes it. A second weak-tier family emits the argmax construction at 27 of 43 valid outputs at the cells where template and argmax differ; the original emits it in 2 of 57. At a held-out cell the original selects the right template and mis-places its one filler. The template is what the model builds reliably, not what it prefers; the library optimizer supplies the execution the weak tier lacks.

And a methods result the table needed. The weak tier's template is a closed form: $k^*(N) = \text{round}(\sqrt{N})$ names the grid. Its value is the empirical modal output at all seven tested N and at four of five held-out N, listed in Supplement S9. Built from zero-shot calls, it does not survive a change of setting. One parent-conditioning step dissolves it and the falsifier fired (the conditioning arm, arm MU). Five generations of a loop do not restore it, and two of its four predictions were not met (the loop arm, arm L). The same prompt sent to the vendor's pinned API yields 0 valid packings in 105, so every prediction there was unscoreable (the pinned-path arm, arm P). The agent runtime yields the template in 4 of 6 in a post hoc control, and extended thinking restores them (the thinking-budget arm, arm P-D). A study that reasons about a zero-shot characterization in a loop owes that step a measurement; here each cost one arm, and the characterization survived none.

Contributions.

1. *The optimizer clears the family; the tiers differ in whether their programs drive it there.* A fixed SLSQP program with no model in the loop clears 45 of 45 and beats the best model-written program at every cell. Among model-written programs, only a Sonnet-tier program with `numpy/scipy` clears the family's best expressible value at rate: 23 of 25 valid programs, or 23 of 41 once every invocation is charged. The isolating arm attributes the gap to the library. Every weak-tier trap cell reads zero under the registered scoring; the post hoc lenient reading recovers 4 of 90 invocations.

2. *The bottleneck is execution.* Handed the argmax, the weak tier builds it in 6 of 14 valid outputs. In code the argmax is reached and never passed; across a vendor, the family that can execute the harder construction emits it; at a held-out cell, the right template gets the wrong filler.

3. *Zero-shot characterization does not transfer.* Three changes of setting each carried a prediction that the anchor persists, and each failed. The falsifier fired under conditioning, the loop missed half its predictions, and pinned-path validity collapsed until the reasoning budget restored it.

Scope. Two tiers of one vendor; a third alias reached 13% validity; the top-alias appendix reports it. Three trap cells, $N = 13$, 21 and 31, carry every channel, and the direct weak-tier row adds $N = 43$. One clearance rule, stated in the notation subsection, carries every clearance verdict. The two serving paths, agent runtime and pinned API, are never pooled. The Sonnet contrast is same-instrument and same-budget since the isolating arm; the weak tier's math-only cells ran at 10 seconds and its library cells at 120, as the limitations section discloses. The direct rows are unconditioned calls; the anchor result is scoped to them.

What is not claimed. Nothing about mechanism inside the weights: the model *emits*, the distribution *concentrates*, the closed form *identifies the modal output*, and the argmax-in-hand arm probes the leading mechanism-free alternative. Nothing about what any named system does at generation zero, or whether its diversity machinery (Mouret & Clune, 2015; Lehman & Stanley, 2011) repairs or explores. Preregistration is an adopted standard (Thomas et al., 2026; Williams et al., 2026; Vaccaro, 2026; Zhang, 2026), not a contribution. Behavioral anchoring is established territory (Huang et al., 2026; Lou & Sun, 2024; Malberg et al., 2025); the anchor here is a construction template, not a number.

2 The family and its argmax

2.1 The benchmark

Maximize the sum of radii of N non-overlapping circles in the unit square. Feasibility is decidable exactly from emitted coordinates; scoring needs no human. The direct-emission prompt (Appendix A) fixes the count, states containment and non-overlap, forbids writing or executing code, and demands a raw Python list of `[x, y, r]` triples. The two program channels replace the code-free clause with a program contract, `math` only or `math` with `numpy` and `scipy`, and score the printed output the same way.

2.2 The recipe family

Three terms: the **recipe family** is the parametric set of grid-plus-filler constructions; a **template** is one member; the **anchor** is the template the distribution concentrates on at a given N.

A $k \times k$ grid places one circle per cell center with $r_{\text{grid}} = 1/(2k)$; m fillers sit on interior grid vertices, tangent to their four surrounding grid circles, with $r_{\text{filler}} = (\sqrt{2}-1)/(2k)$, $0 \le m \le (k-1)^2$. Hence $V(k,m) = k/2 + m(\sqrt{2}-1)/(2k)$. When $N < k^2$ the model does not drop to a smaller grid and fill it; it *truncates*, laying out the $k \times k$ lattice, occupying N cells and keeping the radius, giving $T(k,N) = N/(2k)$. These reproduce every previously reported anchor with no fitting: $V(5,1) = 2.5414214$, $V(5,2) = 2.5828427$, $V(5,0) = 2.500$, $T(5,23) = 2.300$, $V(4,7) = 2.3624369$.

The **family argmax** at N is the largest value the family can express there, recomputed in closed form over every admissible (k,m). Inside a trap zone it is $V(k^*{-}1, m)$, the drop-and-fill construction. An independent linear program, blind to the recipe, recomputes every closed-form value from the coordinates and aborts on disagreement. It covers 83 configurations at $k = 2 \dots 7$ with drift below 10^{-9}, and a further 130 at $k = 8$ and 9 with worst drift 1.8×10^{-14}; Supplement S9.1 lists them. Across the 155 weak-tier rows collected before the extend-cell arm (arm M), only two exceeded the family best, by 3.0×10^{-7} and 3.5×10^{-9}, both below the 10^{-6} bar.

2.3 The order rule is a definition; the branch rule is the hypothesis

We write $k^*(N) = \text{round}(\sqrt{N})$ for the nearest-square order. This is a *definition*, not a hypothesis: two formalizations of "nearest" agree on the integers. The falsifiable content is the **branch rule**: $k^{*2} \le N \to$ extend with $m = N - k^{*2}$ fillers (converge); $k^{*2} > N \to$ truncate the k^*-grid (trap). The extend arm survived testing only at $m \le 1$. A registered extension disconfirmed it at $m \ge 4$, where the model prefers the $(k^*{+}1)$ rectangle grid; Supplement S4 has the modal outputs and the deviation table the verdict. Substituting k^* into the branch condition gives the *trap zones* $N \in [k^2 - k + 1, k^2 - 1]$: $[13, 15]$, $[21, 24]$, $[31, 35]$, $[43, 48]$, $[57, 63]$.

At a converge cell the extend prediction *is* the argmax: we exhausted $N = 10 \dots 80$ and found no exception. The sweep, `extend_branch_scope.py`, finds 38 converge cells, none discriminating, and 27 of 33 trap cells discriminating. So every discriminating cell is a trap cell, and the surviving in-family prediction is the truncate arm $T(k^*, N) = N/(2k^*)$. Every arm tests truncate against drop-and-fill; the governing quantity is the signed distance $N - k^{*2}$, not primality.

2.4 Notation and conventions

Terms used throughout:

Term	Meaning
$k^*(N)$	nearest-square grid order, round($\sqrt{N}$) (a definition, §2.3)
$V(k,m)$, $T(k,N)$	extend-with-m-fillers value $k/2 + m(\sqrt{2}-1)/(2k)$; truncated-grid value $N/(2k)$
anchor	the template the proposal distribution concentrates on at a given N
on-prediction	emitted sum within 2×10^{-3} of the registered predicted value
rival (argmax)	the highest-valued member of the recipe family at that N when it differs from the prediction: inside a trap zone, $V(k^*-1, m)$
discriminating cell	N where prediction $\neq$ family argmax; only these can distinguish anchoring from family search
validity	non-overlap and containment at 10^{-6} (primary) and 10^{-9} (logged), both always reported
trap / converge	branch by sign of $N - k^{*2}$: truncate when $k^{*2} > N$, extend when $k^{*2} \leq N$
tier	nominal capability class of the serving alias addressed (weak/mid/top); `opus_alias` is a bare alias with no attestable weights binding
instrument	the serving path: the agent runtime (decoding parameters not exposed, conditioning context locked only in part) or the vendor's pinned chat-completions API (temperature 1.0, no system prompt); never pooled
UNSCOREABLE	cell with fewer valid samples than its arm's registered floor; claims nothing
evaluability floor	the minimum valid-sample count a cell needs before its verdict counts, fixed per arm at that arm's registration and *not* common across arms: 3 for the GM chain, 5 for arms CN, CP, P, CL, CL-W, CCP and P-D, none for arm CH. Where the loosest one carries a verdict we report the stricter reading beside it
family argmax	the highest value the recipe family can express at N, recomputed in closed form; every clearance verdict is measured against it, and it is the number a discovery claim from a measured proposer has to beat. Table 1 says which proposers clear it
clears at rate	a cell clears *at rate* when its clearance count meets the arm's registered rate bar (20% of valid outputs for arms CL, CCS, CL-W, CCP and B) at every scoreable cell; in plain words, 20% or more of valid outputs clear, at every scoreable cell. A cell with one or two clearances out of dozens shows an *escape*, counted and named, not a clearance at rate
five-valid floor	the evaluability floor of 5, the one arms CN, CP, P, CL, CL-W, CCP and P-D carry (the evaluability-floor row above): a cell with fewer than five valid outputs is not scoreable on its own; pooled counts are reported beside it
dead zone	clearance above zero and under the 20% bar: a count that is neither the registered zero nor a clearance at rate
template anchor $T(k^*, N)$	the truncated-grid value at the nearest-square order, $N/(2k^*)$, the anchor of the truncate branch (§2.3); at a trap cell it sits below the family argmax
grid order k	the k of a $k \times k$ grid, one circle per cell centre at $r = 1/(2k)$; k^* is the nearest-square order, and a packing's grid order is backed out of its dominant emitted radius (Supplement S9.2)

The clearance rule, stated once. Five thresholds appear: 10^{-6} and 10^{-9} for validity, 2×10^{-3} for value matching, 10^{-6} again as a minimum excess, and 5×10^{-8} as the granularity at which sums are recorded. One rule governs every claim that an output *left the family*:

An output leaves the family iff it is valid at 10^{-9} *and* its sum exceeds the family argmax by more than 10^{-6}.

Each condition excludes a real set of rows. The validity condition excludes eighteen rows of the conditioning arm (arm MU), a post hoc re-read. They sit above the argmax at the primary tolerance by at most 1.5×10^{-6} yet fail the tighter gate; their excess is their overlap. The excess condition excludes the two square-arm rows above, at 3.0×10^{-7} and 3.5×10^{-9} over the family best and at the argmax as recorded. Across the arms swept before the library arm (arm CL), exactly three rows leave the family. Two are converge-cell rows of the extend-cell arm, from a post hoc sweep; one is a Sonnet sample at $N = 31$; all three are out-of-family constructions. The library arm's Sonnet tier adds twenty-three more, and a post hoc lenient reading of the weak tier's library programs four more that the registered parser does not count. The 2×10^{-3} value window never carries a clearance verdict; §3.3 records the one case where it was mistakenly allowed to.

3 Main result: who clears the family argmax

Twenty-six registered arms appear in this paper, each reported whichever way it went, and the falsifiers that fired are flagged in place. Every prediction whose verdict this section reports was registered before the arm's first scored run, with one exception. The Sonnet direct arm's registration named 5 of 20 samples at $N = 13$ and 21 as already seen, with $N = 31$ fully blind. The weak math-only replication and the Sonnet math-only arm were registered after the first weak math-only arm's results were known and before their own sampling. Readings marked post hoc were not registered. Commits, amendments and outcomes are in Table 5. Table 2 lists the eleven arms that carry the channel table, the optimizer-alone baseline among them; all twenty-six are in Appendix B. The remaining arms establish the mechanism, the transfer boundaries or the mode prediction of Supplement S9. The trace arm (arm T), the rectangle arm (arm G) and the top-alias arm (arm O) carry no verdict the table needs; they are reported in the top-alias appendix and the supplement. Arms whose verdicts rest on non-discriminating cells, on samples under their floor, or on a null that does not separate have a row in the appendix ledger (Table 4) and full text in the supplement. The reproducibility section states that criterion.

Table 2: The eleven arms that carry Table 1, with the optimizer-alone baseline: section (a letter is an appendix; Sn is a section of the supplement), proposer, design, registered test, outcome. All twenty-six registered arms are in Appendix B (Table 4).

Arm	§	Proposer(s)	Design	Registered test	Outcome
F (square, bare)	3.3, S9	weak tier	7 N × 5–20	P1–P5 exact-value modes	modal at 7/7 N
S (Sonnet direct arm)	3.3	Sonnet tier	3 N × 10	P-S1–P-S4 tier contrasts	1/30 on-prediction, 100%/90% valid ($10^{-6}/10^{-9}$); one out-of-family escape; 5 of 20 samples at $N = 13/21$ seen before registration, disclosed
M (extend-cell arm)	3.3, S4	weak tier	4 N × 15	P-M1–P-M4 + F-M1/F-M2	F-M1 **triggered 3/3**: extend branch dies at $m \geq 4$; fifth k confirms; two out-of-family escapes at $N = 20$ (post hoc)
CC (weak math-only arm)	3.4	weak tier	3 N × 15	P-CC1 vs P-CC2	P-CC1 holds: anchor stays modal; 0/37 above the family argmax
CC2 (replication arm)	3.4	weak tier	3 N × 15	R1 + R2 + F-CC2.1	**REPLICATED**: 0/41 above argmax, anchor modal 3/3
CCS (Sonnet math-only arm)	3.4	Sonnet tier	3 N × 15	P-CCS1 vs P-CCS2	P-CCS2 holds: 0/37 above argmax — the channel, not the tier
P (pinned path)	5, S8	weak tier, bare API	15 cells × 15	P-P1–P-P4 + F-P1	validity collapses: 0/105 valid at the square cells, 4/75 held-out; every direct-emission prediction UNSCOREABLE; P-P3 holds (0 of 12 valid programs clear); same-day agent-runtime control 6/6 valid, anchor 4/6 (post hoc)
CL (library code)	3.5	weak + Sonnet, numpy/scipy	2 tiers × 3 N × 15	P-CL1 vs P-CL2 + F-CL1	weak: 0/11 clear, P-CL1 holds; Sonnet: **23/25 clear at 3/3 cells**, P-CL2 holds, **F-CL1 fires**

Arm	§	Proposer(s)	Design	Registered test	Outcome
CCP (isolating arm)	3.5	Sonnet tier, bare API, math only, 120 s	3 N × 15	P-CCP1 vs P-CCP2 + F-CCP1	P-CCP1 holds: 0/35 clear, none reaches the argmax; best per cell 1.625, 2.1, 2.673 — the library, not the path or the budget
CL-W (weak top-up)	3.5	weak tier, numpy/scipy	3 N × 15, pooled with CL	P-CLW1 vs P-CLW2 + F-CLW1; lenient reading secondary	P-CLW1 holds: 0/19 pooled clear, upper bound 16.8%; lenient reading 4/41, dead zone
B (optimizer alone)	3.5, S7	no model in the loop; a fixed SLSQP program	3 N × 15 seeds, 120 s	P-B1 vs P-B2 + S-B1	P-B1 holds: **45/45 clear**, 3/3 cells; S-B1: beats the Sonnet best at 3/3 cells; amendment 1 (version 1 blocked by the allowlist, kept)

3.1 Registration protocol

For each cell the prompt is pinned and its SHA-256 digest written to disk first; the $N = 13$ square prompt hashes to `32db485b...`. Predictions live in the header of each arm's analysis script or in a standalone registration file. Competing predictions, numeric thresholds, tie conventions and a falsifier are fixed before the first invocation. Raw outputs are stored verbatim, failures included; scoring is deterministic and local. Supplement S9.2 records which of the bare square arm's seven cells carry an individually registered point prediction and which follow from the same closed form without one.

Four properties are disclosed, not repaired. Temperature, top-p and top-k are not exposed by the agent runtime. The alias-to-weights binding is a promise, not a hash. The subagent inherits instruction files outside the task prompt, so the digests lock a *fragment* of the conditioning context, and a replicator cannot reconstruct the inherited files from what we release. Hash coverage is incomplete, as the reproducibility section states. The pinned-path arm (arm P) and the library arm (arm CL) exist because of the first and third. They send registered prompts through a serving path that exposes and pins temperature and carries no inherited context.

Two scoring conventions were fixed in advance. Validity is reported at 10^{-9} and 10^{-6}, both logged, 10^{-6} primary. Proposers print six to eight decimals, so an eight-decimal tangency misses contact by $\sim 5 \times 10^{-9}$, while 10^{-6} sits far below the $\sim 10^{-2}$ gap between rival constructions. Value matching uses a 2×10^{-3} window, which never carries a clearance verdict. Every on-prediction rate is computed over *valid* outputs and so conditions on execution success, the variable the argmax-in-hand arm (arm CH) shows can separate what a model attempts from what it lands. Per-cell validity denominators are reported beside every rate so the reader can bound the exposure.

3.2 The table

Table 1 is where the tiers and channels are scored. Each row is one arm, or one registered pair of arms, and "clear" counts valid outputs under the clearance rule of §2.4. Five readings follow.

One cell clears at rate. Sonnet-tier programs allowed `numpy` and `scipy` clear the family in 23 of 25 valid cases, at 3 of 3 cells. Their best program per cell beats the argmax by 2.5%, 3.6% and 4.2%, and their median clearing program by 3.6%. They sit above the grid family, not at the frontier: against the published best-known sums the best programs are 0.5%, 0.9% and 0.8% below at $N = 13$, 21 and 31. A fixed reference program with no model in the loop, run through the same pipeline, clears 45 of 45 and beats the best Sonnet program at every cell (arm B). The Sonnet row clears because its programs invoke that optimizer.

Neither tier without the optimizer; the optimizer without any tier. Weak-tier programs with the same libraries clear 0 of 11 (arm CL). Pooled with their top-up (arm CL-W) they clear 0 of 19, and the 95% Wilson upper bound of 16.8% excludes the 20% bar. Two caveats. The $N = 21$ cell holds only 3 valid outputs, under the five-valid floor. A post hoc reparse that accepts `numpy` scalar output reads 4 of 41, a point estimate of 9.8% whose upper bound of 22.5% does not exclude the bar. The weak tier clears at rate under neither reading, and the exclusion belongs to the registered reading alone. Sonnet-tier programs without the libraries clear 0 of 37 with simulated annealing over hand-rolled generators, Halton multi-restarts and force-directed relaxation. On the library row's own serving path and 120-second budget they clear 0 of 35 without reaching the argmax (arm CCP, the isolating arm). Every weak-tier cell is 0 at the trap cells under the registered scoring. The optimizer-alone baseline that clears at every cell calls the same library the weak tier's 19 valid programs call.

Escapes exist and are counted. Two cells show low-rate escapes, both out-of-family constructions found by sampling rather than by in-family search. One Sonnet direct emission in 30 clears at $N = 31$. Two weak-tier emissions in 15 clear at the extend cell $N = 20$, found in a post hoc sweep. They are in the table because a zero that hides them would be false. They do not change which cell clears at rate.

Rows are never pooled across serving paths. The same weak-tier prompt yields 78% valid packings through the agent runtime and 0 of 105 through the pinned API (arm P). The clearing cell ran on the pinned path. The Sonnet math-only cell ran on the runtime, under a 10-second execution budget against the library arm's 120 seconds. Within the weak tier the library comparison is same-path: 0 of 12 math-only and 0 of 11 with libraries, both pinned, and 0 of 19 with the top-up. Each of those two arms has one cell above the five-valid floor, $N = 21$ for math-only and $N = 31$ for libraries, and both read 0 of 6 there. That comparison still crosses the budget, 10 seconds against 120. Within the Sonnet tier the isolating arm removes the crossing: the weak math-only arms' prompt on the pinned path under 120 seconds clears 0 of 35. The library's credit is that same-instrument gap, 0 of 35 against 23 of 25, on one path and one budget; the runtime figure of 0 of 37 crosses instruments and attributes nothing.

The same rows, per invocation. Every rate above conditions on validity, and validity is what the manipulation moves, so Table 3 also charges every invocation, valid or not. That reading changes no registered verdict. The Sonnet tier clears 23 of 41 invocations, the weak tier 0 of 90, or 4 of 90 under the lenient reparse, and the baseline 45 of 45. The Sonnet denominator is 41 because the serving path refused four invocations with an HTTP 402, and a refusal is charged to the path. The lenient weak reading stays below the 20% bar on this count too, at a 95% Wilson upper bound of 10.9%.

Outside the table. Three arms sample regimes no row covers: five held-out cells absent from every scoreboard (arm CN), one parent-conditioning step (arm MU) and five generations of a proposer loop (arm L). None of their valid outputs clears the family argmax either; for arm MU that is a post hoc re-read, since no clearance prediction was registered for it. Pooled with the three math-only program rows, 0 of 290 valid outputs across six arms clear. That pool counts arm MU at 10^{-9} and the other five arms at the primary tolerance. Held at 10^{-9} throughout it reads 0 of 281. Held at 10^{-6} throughout it reads 315 valid outputs with 18 above the argmax, every one of the 18 an arm-MU overlap that the tighter gate rejects. The pool, its membership rule and its tolerance accounting are in Supplement S1. The pool describes those arms and is not a bound: the escapes above sit outside it by named regime.

3.3 Direct emission

The weak tier emits a computable template. The bare square arm (arm F) produces uniform grids of radius $1/(2k)$ truncated to N circles, with interior-vertex fillers only when the grid underfills. Its modal output is the closed form's prediction at all seven tested N. At the four discriminating cells, $N = 13$, 21, 31 and 43, where prediction and family argmax differ, 41 of 57 valid emissions land on the prediction and 2 reach the rival argmax, both at $N = 21$. No valid emission at those cells clears the family. Supplement S9 holds the per-cell table, the template-shape null, the held-out replication at five N and the container and paraphrase probes.

Two weak-tier emissions do clear the family, the direct row's second entry. At the extend cell $N = 20$ the extend-cell arm (arm M), reported in Supplement S4, disconfirmed the registered prediction $V(4, 4) = 2.2071068$: the model emits the 5×4 rectangle, $T(5, 20) = 2.0$, in 12 of 15. Two of the fifteen valid rows are staggered rows of four at $r = 1/9$, sum 2.2222222, a hexagonal layout the family does not contain, valid at 10^{-9}. They are not the modal output, they were found post hoc, and they are counted.

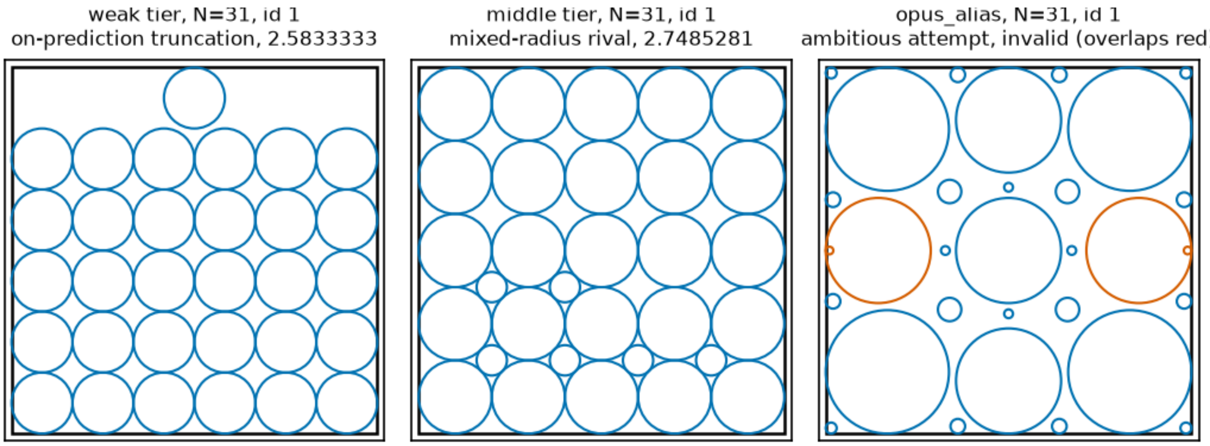

Figure 1: *Three attractor families at a single cell,* $N = 31$. One representative packing per tier, ledger sample id in each panel title (weak tier `bare` id 1, on-prediction truncation 2.5833333; Sonnet tier `sonnet_bare` id 1, mixed-radius rival 2.7485281; `opus_alias` id 1, invalid, with only the offending circles highlighted). Source: `arm_f_candidates_v2.jsonl` via `fig_scripts.py`, deterministic.

The Sonnet tier perturbs and mixes. The Sonnet direct arm (arm S) was valid 30/30 at 10^{-6} and 27/30 at 10^{-9}, and almost never on-prediction: 0/10, 0/10, 1/10, against the weak tier's 12/16 at the same cells. Instead of truncating a uniform grid it perturbs one, with enlarged edge rows, hexagonal interior rows and two or three distinct radii in the same packing, 29/30 multi-radius, as Figure 1 shows. One sample at $N = 31$ emitted 27 circles at $r = 1/12$ with 4 corner circles at $r = 1/8$, summing to 2.7499999991, above the family's 2.7485281. Recomputed from its coordinates the construction is tangency-tight, with minimum pairwise and wall slack exactly zero, and it is among the 27 samples valid at 10^{-9}. It is the direct-emission row's one clearance: 1 of 30 at the primary tolerance, 1 of 27 at 10^{-9}.

That sample also exposed a scoring defect. At $N = 31$ the family rival, 2.7485281, and this out-of-family value, 2.75, differ by 1.47×10^{-3}, inside the registered 2×10^{-3} window, so the value classifier counted the escape as a hit on the construction it exceeded. The window was chosen for the anchor comparison, where the nearest competing value is $\sim$0.15 away. Classified by structure instead, the Sonnet rival count at $N = 31$ is 2/10 and pooled 5/30. A third alias, addressed through a bare tier name we call `opus_alias`, is the top-alias arm (arm O): valid in 4 of 30, and reported in Appendix D with the ladder it completes.

3.4 Math-only programs: an executed program does not escape the family

No math-only program clears the family at either tier. The weak math-only arms (arms CC and CC2) put 0 of 78 valid programs above the family argmax and exactly two on it; the Sonnet math-only arm (arm CCS) puts 0 of 37 above it. The channel changes dispersion, not escape. Deployed loops elicit *programs*, and the code-free design had assumed a code channel would route construction to an executed optimizer; the first weak math-only arm measures that assumption. It ran the three trap cells $N = 13, 21, 31$ at $n = 15$ weak-tier invocations per cell. The prompt is the bare stem with exactly two registered line replacements: the code-free clause becomes a program contract, standard-library `math` only, and the output clause asks for program source. Programs run sandboxed under `python -I` with a 10-second timeout and a registered AST allowlist; the executor and its four-bin smoke test sit in the registration commit. Stdout is scored by the bare square arm's conventions unchanged. Two predictions competed: the modal valid executed output remains the $T(k^*, N)$ bucket at 2 or more of 3 cells, or it sits strictly above the anchor at 2 or more of 3. Ties or mixed modes count as neither.

N	Valid	Modal bucket (count, margin)	Anchor-rate	Above anchor
13	12/15	anchor 1.6250000 (2/12, one sample)	2/12	2 — 1.6614054, and the family argmax exactly
21	13/15	anchor 2.1000000 (9/13, margin 8)	9/13	0
31	12/15	anchor 2.5833333 (3/12, one sample)	3/12	0

The anchor stays modal at 3 of 3 cells, on thin margins. At $N = 21$ the mode is heavy: nine exact 2.1000000 emissions, most programs writing the 5×5 grid truncated to 21 circles. At $N = 13$ and $N = 31$ the modal bucket is the anchor value with a one-sample margin over dispersed sub-anchor values, so the modal-identity claim is thin at two of three cells. The robust result is the registered decomposition row: no valid program output exceeds the family argmax anywhere, 0 of 37 above the rival. Only two exceed the anchor at all, both at $N = 13$: one at 1.6614054, and one program that computes the family argmax construction $V(3,4) = 1.7761424$ *exactly*. Of the 45 programs, 7 produce geometrically invalid packings, 1 imports `random` against the stated contract and is caught by the registered gate, and none times out. Code-free anchor concentration at these cells runs 56 to 80%, the code channel's pooled anchor rate is 14/37, and the lost mass sits entirely *below* the anchor: greedy radius-growth heuristics converging under the lattice.

Both registered follow-ups hold, and they harden the result. A fresh-draw replication wave (arm CC2) and the Sonnet math-only arm (arm CCS) were registered together, after the first wave's results were known, 45 invocations each on the byte-identical prompts. The replication returned REPLICATED under its pre-stated verdict map: 0 of 41 valid outputs above the rival, exactly the registered zero-count prediction, and the anchor bucket modal at 3 of 3 cells. That includes $N = 31$, where the mode that was one-sample-thin in the first wave arrived at 10 of 15 valid with a margin of 8. The two pooled programs that reach the argmax come one per wave, both at $N = 13$. The Sonnet math-only arm's zero-count prediction held against a registered escape at 20% or more above the rival: 0 of 37 valid Sonnet-tier outputs exceed the argmax. Its programs are qualitatively different search, simulated annealing over hand-rolled linear-congruential generators, Halton-sequence multi-restarts and force-directed relaxation, against the weak tier's greedy radius growth. The tier barely anchors, 5 of 37 on the anchor value, yet its executed math-only search never passes the family argmax at any cell. The replication returned 41 valid, 3 geometry-invalid and 1 blocked import; the Sonnet arm 37 valid, 7 geometry-invalid and 1 timeout. One replication invocation, $N = 13$ slot 2, used runtime tools despite the no-tools wrapper; it is scored as emitted, and the deviation table records it. Ledgers and scripts for these arms are listed in Appendix E.

3.5 Library programs: the channel with numpy and scipy

A Sonnet-tier program allowed `numpy` and `scipy` clears the family in 23 of 25 valid cases, at 3 of 3 cells. A fixed `scipy.optimize` program with no model in the loop clears 45 of 45 through the same pipeline. A Sonnet-tier math-only program on the same path and budget clears 0 of 35 and never reaches the argmax. A weak-tier program with the same libraries clears 0 of 19 under the registered parser, and its clearance rate is bounded under the 20% bar; a post hoc reparse reads 4 of 41. The clearing is the optimizer's: the Sonnet tier clears because its programs drive that optimizer, and a weak-tier program calling the same optimizer does not clear at rate. Table 3 holds every per-cell count, best sum and prediction verdict.

The arm. The library arm is the weak math-only arms' prompt with one clause changed: the import allowlist widens from `math` to `math`, `numpy` and `scipy`, the channel deployed loops permit. It ran at two tiers, the vendor's `claude-haiku-4.5` and `claude-sonnet-4.5` aliases, at temperature 1.0, on the pinned path of the pinned-path arm. Each tier sampled the three trap cells at $n = 15$ per cell, 90 invocations in all. The serving path refused 4 of them, all Sonnet-tier, with an HTTP 402, recorded as refusals; 85 returned a parseable program and one Sonnet response failed to parse. Each program runs in a fresh `python -I -S` subprocess under a 120-second wall clock, scored by the bare square arm's conventions, with clearance decided only by the clearance rule. The child runs with `-I`, which ignores `PYTHONPATH`, so a fixed driver, identical for every row and outside the scored source, inserts the interpreter's site-packages into `sys.path` and runs the model's source unmodified through `runpy`. That correction was made before any sampling; the deviation table records it. Each tier carried two competing predictions, no valid output clears against at least 20% clearing. The Sonnet tier also carried a falsifier, clearance at two or more of the three cells, with its integration rule written down in advance.

Arm, reading	$N = 13$	$N = 21$	$N = 31$	Pooled; prediction verdicts
Clear of valid, by cell				
weak library (CL)	0 of 3	0 of 2	0 of 6	0 of 11; Wilson upper 25.9% pooled, 39.0% at $N = 31$; P-CL1 holds, both bounds above P-CL2's 20% bar
weak library (CL), lenient, post hoc	3 recovered, 1 clears	4 recovered	2 recovered	1 of 20; dead zone
weak top-up (CL-W), new rows	3 valid	1 valid	4 valid	8 valid of 45
weak pooled (CL + CL-W)	0 of 6	0 of 3	0 of 10	0 of 19; Wilson upper 16.8%, under the 20% bar; P-CLW1 holds, F-CLW1 does not fire
weak pooled, lenient, post hoc	2 clear	1 clears	1 clears	4 of 41, 9.8%, upper bound 22.5%; dead zone
Sonnet library (CL)	7 of 9	11 of 11	5 of 5	23 of 25; P-CL2 holds, F-CL1 fires at 3 of 3 cells
isolating arm (CCP)	0 of 10	0 of 14	0 of 11	0 of 35; P-CCP1 holds over P-CCP2, F-CCP1 does not fire
optimizer-alone baseline (B)	15 of 15	15 of 15	15 of 15	45 of 45; P-B1 holds over P-B2, S-B1 holds at 3 of 3 cells
Clear per invocation, every call charged				
weak pooled (CL + CL-W)	0 of 30	0 of 30	0 of 30	0 of 90; Wilson upper 4.1%
weak pooled, lenient, post hoc	2 of 30	1 of 30	1 of 30	4 of 90; Wilson upper 10.9%, under the 20% bar
Sonnet library (CL)	7 of 15	11 of 14	5 of 12	23 of 41; 4 HTTP 402 refusals charged to the path
optimizer-alone baseline (B)	15 of 15	15 of 15	15 of 15	45 of 45
Best sum per cell				
family argmax, closed form	1.776142375	2.258883476	2.748528137	the bar a clearing sum exceeds by more than 10^{-6}
published best-known, lower bound	1.829	2.362	2.889	three truncated decimals, from the bound table
Sonnet library (CL)	1.820699211	2.340549845	2.864990189	2.5%, 3.6% and 4.2% above the argmax; 0.5%, 0.9% and 0.8% below best-known
isolating arm (CCP)	1.625	2.1	2.672523443	all under the argmax; at $N = 13$ and 21 the anchor $T(k^*, N)$
optimizer-alone baseline (B)	1.829542412	2.359585843	2.883035274	above the Sonnet best at 3 of 3 cells; 1.8295 matches 1.829+ at $N = 13$; 0.1% and 0.2% below best-known at $N = 21$ and 31
weak pooled, lenient clearing sums	1.784872559, 1.812531608	2.331990574	2.851728045	post hoc; the 1.784872559 program is unseeded

Table 3: **Library-channel arms, cell by cell.** Every count, best sum and prediction verdict of this subsection. "Clear" counts valid outputs above the family argmax under the clearance rule. Wilson bounds are 95% upper bounds on the clearance rate. Lenient rows re-read the same outputs with a parser that accepts `numpy` scalars. For arm CL that reparse is post hoc and carries no registered verdict; for the top-up (CL-W) the lenient reading was fixed in its registration as the secondary reading, which the pooled lenient row reports. The per-invocation block charges every call, valid or not, and carries no registered verdict of its own: the registered readings are the clearance rates over valid outputs above it.

Weak tier: none clears, underpowered. The weak tier returned 11 valid programs of 45, every one calling `scipy.optimize`, and none clears the family. The 34 failures are 22 stdouts the registered parser could not read, 7 invalid packings, 4 execution errors and 1 blocked import; none timed out. The no-clearance prediction holds. Only $N = 31$ is above the five-valid floor: 0 of 6 there and 0 of 11 pooled. The 95% Wilson upper bounds are 39.0% and 25.9%, both above the 20% bar the rival prediction set. The arm shows the weak tier did not clear, not that it cannot clear at rate; the top-up below powers the cell.

Post hoc: what the 22 unreadable outputs contain. All 22 parse failures print their packing as `numpy` 2.x scalar reprs, `np.float64(0.49...)`, which the registered parser's `literal_eval` rejects. No Sonnet-tier program did this. Stripping that wrapper, and nothing else, parses all 22 over a re-execution that reproduced every registered bin. Eleven of the 22 call `numpy.random` unseeded, so their sums belong to that execution. Of the 22, 9 are valid at both tolerances. The other 13 are invalid packings: 7 overlaps, 4 outside the square and 2 with non-positive radii. One clears, a deterministic $N = 13$ program with 13 distinct radii summing to 1.812531608, 2.0% above the argmax. Under this reading the weak tier is 1 of 20 pooled, the 20 being its 11 registered-valid programs plus the 9 recovered here, and 1 of 6 at $N = 13$, a rate in the dead zone. The registered count stands as the arm's result; the lenient count is reported beside it because a zero that depends on a print format is the wrong thing to lean on.

The weak-tier top-up (arm CL-W). The top-up was registered after the reparse, with the pooling rule, both readings and its predictions fixed in advance. It reruns the weak-tier prompt byte-identical, same path and decoding, for 45 more invocations, 15 per cell; the registration drafted 90, the credit allowed 45, and the deviation table records the cut. The 90 rows of both runs are scored in one pass by the library arm's pipeline. Of the 45 new rows 8 are valid. The failures are 23 unreadable stdouts, again `numpy` scalars, 4 invalid packings, 3 execution errors, 3 blocked imports, 3 timeouts and 1 with no program. Under the registered parser the pooled weak tier holds 6, 3 and 10 by cell, all valid at 10^{-9}, and 0 of 19 valid clear. The $N = 21$ cell stays under the five-valid floor and claims nothing on its own. The 95% Wilson upper bound on the pooled count is 16.8%, so the weak library cell now excludes the 20% bar. The no-clearance prediction holds and the falsifier does not fire. Under the post hoc lenient reading 41 are valid and 4 clear, 9.8% with an upper bound of 22.5%, inside the dead zone. The last of those four sums, 2.851728045 at $N = 31$, is a sum the Sonnet tier also reached. An independent re-scoring with its own parser reproduced every bin and three of the four sums to the digit; the 1.784872559 program is unseeded, and its sum moves with the draw. The weak tier does not clear at rate under either reading.

Sonnet tier: 23 of 25 clear. The Sonnet tier returned 25 valid programs of 45 at the primary 10^{-6} tolerance, every one calling `scipy.optimize`. Of those 25, 23 clear the family, and each of the 23 is valid at 10^{-9} as the clearance rule requires. The two non-clearing programs, both at $N = 13$, are also valid at 10^{-9}; their sums are 1.5 and 1.767686476, under the argmax. The count is therefore 23 of 25 at either tolerance. The best program per cell exceeds the argmax by 2.5%, 3.6% and 4.2%. Across the 23 clearing programs the excess runs from 0.67% to 4.24%, with a median of 3.62%. The 20% prediction holds and the falsifier fires at 3 of 3 cells. Against the published best-known values, three-decimal lower bounds from the bound table of Supplement S9.1, the best programs sit 0.5%, 0.9% and 0.8% below.

The isolating arm (arm CCP). Between the Sonnet math-only cell at 0 of 37 and the library cell at 23 of 25, three things changed along with the import allowlist: the serving path, the dispatch context and the execution budget. The math-only arm CCS ran on the agent runtime under a 10-second budget; the library arm ran on the pinned API under 120 seconds. Neither cell alone said which factor cleared. The isolating arm reruns the weak math-only arms' prompt byte-identical on the library arm's path: Sonnet tier, pinned API, temperature 1.0, no system prompt, a 120-second budget, `numpy` and `scipy` blocked by the AST gate. Its competing predictions were no clearance, meaning the library is what clears, against clearance at 20% or more at two of three cells, meaning path or budget explain the library arm. Of 45 invocations, one was refused in flight at row 24 and resumed with one worker, request unchanged. Valid at both tolerances are 35 programs, 10, 14 and 11 by cell, and 0 clear; 10 are geometrically invalid, with no timeouts and no blocked imports. None reaches the argmax, and at $N = 13$ and $N = 21$ the best program is the template anchor $T(k^*, N)$ itself. The anchor's structure is modal: 7 of 10 programs at $k = 4$ and 12 of 14 at $k = 5$. The no-clearance prediction holds and the falsifier does not fire. The three Sonnet program cells now read 0 of 37, 0 of 35 and 23 of 25, and only the third had the libraries. Within the Sonnet tier, path and budget moved nothing and the libraries moved everything. At the weak tier the same libraries move nothing at rate. Leaving the family at rate took the tier *and* the optimizer.

The optimizer alone (arm B). The control the title asks for is a fixed `scipy.optimize` program with no model. The optimizer-alone baseline was registered before its first scored run; Table 5 records its post-review motivation. Its competing predictions: the program clears at 20% or more of valid runs at two or more cells, or it never clears. Under the first, the optimizer alone clears and contribution 1 is rewritten by a rule fixed in the registration. A secondary prediction asked whether its best packing per cell exceeds the Sonnet best. The program is random-restart SLSQP over centre and radius variables with analytic gradients, uniform random starts, a uniform radius shrink by the largest constraint violation, and 50 restarts. Its 94 lines are in Supplement S7; Claude drafted it under the human author, and no result informed either version. "No model" means no model call at proposal time: one fixed program serves every N and seed, where every other program cell in the channel table is written afresh by a model at each invocation. It ran unmodified through the library arm's pipeline and scoring, on one core, at the three cells with 15 seeds each. Version 1 bounded its restarts by wall clock, importing `time` and `sys`, which the allowlist forbids; the AST gate blocked all 45 rows, and that run is kept. Version 2 fixes the count at 50, a margin under the 80 restarts an unscored timing run fit in 95 seconds at $N = 31$, and changes nothing else. Result: 45 of 45 valid at both tolerances and 45 clear, 15 of 15 at every cell. Its best sum per cell beats the Sonnet best at 3 of 3 cells, so the secondary prediction holds. At $N = 13$ its 1.8295 matches the published 1.829+ to the three truncated digits published. Its packings are off-family: dominant grid orders 3 to 7 across cells, no single template. The first prediction holds and the integration rule applies: the library optimizer clears the family with no model in the loop.

What the arms establish. Within the Sonnet tier the library is the factor, and with the optimizer-alone baseline the library is sufficient. A weak-tier program invokes the same library in every one of its 19 valid programs and clears nothing at rate, and no math-only program clears at either tier, on either path or at either budget. The tiers differ in whether their programs drive the optimizer to a clearing packing, the execution finding the mechanism section reports for the code channel. The 10-second budget of arms CC, CC2 and CCS remains a difference from the library arm for the weak tier's own contrast. Per the registered integration rules, the abstract and title name the library-enabled Sonnet tier as the regime that clears. The isolating arm's 35 valid programs are reported by name, not pooled with the 0 of 37. Ledgers and scripts are listed in Appendix E.

4 Mechanism: choice follows the score table, execution does not

Table 1 says a weak-tier proposer leaves the family at rate in no channel, and a Sonnet-tier one leaves it at rate only in the library cell. Two readings of that are open: the weak tier *prefers* the template, or it *cannot build* anything else. One registered arm separates them, and three other directions show the same split. The predictions whose verdicts this section reports were registered before their arm's first scored run; the readings marked post hoc were not. Their outcomes are in Appendix B, and every post-hoc reading below has a row in the deviation table.

4.1 Arm CH: the argmax in hand does not dislodge the template

Handed the argmax, the model still emits the template, yet reaches for the argmax. Under the registered scorer the template prediction held at 2 of 3 cells. Counted at the attempt level, post hoc, the model reaches for the argmax in 36 of 45 invocations and builds it in 6 of 14 valid outputs.

The argmax-in-hand arm (CH) hands the model the bare prompt plus every admissible $T(k, N)$ and $V(k, m)$ value, the family argmax included, and tells it to use whichever scores highest or anything better. It ran 15 invocations per cell at $N = 13$, 21 and 31, 45 in all, registered together with the conditioning arm (MU) before sampling. The tractability alternative of the limitations section says the model truncates at k^* because it can compute nothing else. Enumerating the family removes that burden, so under it the modal output should become the stated argmax; under the template reading it stays $T(k^*, N)$.

The registered verdict carries a survivorship artifact, reported rather than exploited. The registered scorer takes the modal output among *valid* samples, and under it the template reading wins. At $N = 13$ the modal valid output is $T(4, 13) = 1.6250000$ against a stated argmax of 1.7761424. At $N = 31$ it is $T(6, 31) \approx 2.5833$, 4 of 7 valid, against 2.7485281 at 3 of 7: a one-sample margin. Only $N = 21$ goes to the argmax, 2.2588835 in 3 of 5 valid, so the argmax prediction held at 1 of 3 cells. This arm set no floor of its own and grants $N = 13$ a verdict on two valid samples. Under the GM chain's floor of three that cell would not be scoreable, and the template prediction would hold at 1 of 2. Both readings are reported, since each floor was fixed at its own arm's registration (§2.4). But validity is low, 2, 5 and 7 valid of 15 by cell, and the failure modes change the picture at the *attempt* level.

The attempt-level reading, post hoc. Of the 31 invalid rows, 30 are evaluable. **All 30 attempt the argmax at its own grid order** k^*-1: 12 of 13 at $N = 13$, 10 of 10 at $N = 21$ and 8 of 8 at $N = 31$. The order is backed out of each row's dominant emitted radius, as Supplement S9.2 recovers k. The thirty-first row emits radical literals the scorer cannot evaluate and is not scored as attempting anything. All 30 misplace the fillers, typically at the container corners, where radius $(\sqrt{2}-1)/(2k)$ overlaps the adjacent grid circle instead of sitting in a grid interstice.

The control this reading needs. Attempt counts need the unprompted rate. The bare square arm (F) supplies it: its invocations at the same three cells, under the same dominant-radius estimator. Over *all* evaluable rows, valid and invalid alike, the bare square arm reaches for the argmax order in **5 of 57**, or 9%, against the argmax-in-hand arm's **36 of 44**, or 82%. Its one-sided Fisher p, post hoc and uncorrected, is in Table 5 (`arm_f_attempt_uncond.py`). Those 57 are the bare square arm's evaluable rows at $N = 13$, 21 and 31, not the 57 valid emissions of the direct-emission subsection's four cells. The invalid-row comparison conditions on a variable the manipulation itself moves, a collider: the argmax-in-hand arm is 31 of 45 invalid and the bare square arm 10 of 60 at the same cells. For the record it points the same way, 3 of 7 evaluable invalid bare rows at the rival order against 30 of 30 (`arm_f_attempt_control.py`); its p is in the same table. Both comparisons are post-hoc strata, not registered outcomes.

What the arm supports. The 36 of 45 attempts are the 30 evaluable invalid attempts plus 6 valid outputs that land the argmax. The control quotes 36 of 44 because the forty-fifth row, the radical-literal one, is not evaluable and leaves both arms' denominators there. The model builds the argmax in 0 of 2 valid at $N = 13$, 3 of 5 at $N = 21$ and 3 of 7 at $N = 31$, 6 of 14 pooled. It does not *fail* to build the argmax; it builds it unreliably. The registered metric conditions on execution success, the very variable that separates the two readings, so the summary is a decomposition rather than a verdict. Handed the argmax, the model's *choice* follows the score table. Its *execution* succeeds less than half the time it is attempted, failing off-template in 30 of the 31 invalid attempts on the post-hoc reading. The tractability alternative is wrong about selection and right about instantiation.

N	Stated argmax	Modal valid output	Valid	Invalid rows attempting argmax	Registered verdict
13	1.7761424	$T(4,13) = 1.6250000$	2/15	all	P-CH1 holds
21	2.2588835	argmax, 3/5	5/15	all	P-CH2 (argmax)
31	2.7485281	$T(6,31) \approx 2.5833$ (4/7)	7/15	all	P-CH1 holds (one-sample margin)
pooled	n/a	template 2 of 3 cells	14/45	**30/30** evaluable (all misplace fillers; 1 further row unevaluable, radicals)	choice follows score table (36/45 attempt it); execution lands it 6/14 valid

4.2 The same asymmetry from three other directions

In code. An executed math-only program is the cleanest removal of the hand-computation burden. It recovers the argmax construction exactly twice in 78 valid weak-tier programs, one per wave, both at $N = 13$ and each computing $V(3,4) = 1.7761424$, and it never passes it (the math-only subsection). With `scipy.optimize` available the weak tier clears nothing in 19 valid programs under the registered parser, and 4 of 41 under a post-hoc lenient one. The Sonnet tier clears 23 of 25, and 0 of 35 without the libraries on the same path and budget. Every valid weak-tier library program calls `scipy.optimize`, and none drives it past the family. The library supplies execution the weak tier invokes but does not steer to a clearing packing; the optimizer-alone baseline (B), a fixed program with no model, steers it there at 45 of 45.

At a held-out cell. The held-out-cells arm (CN) has one miss, $N = 50$, and it is the argmax-in-hand pattern in the wild. There 10 of 11 valid outputs are the registered 7×7 grid plus exactly one filler, and 6 of them push the filler into a container corner instead of the interstice. The rows are Supplement S9.4. The template is selected; the filler is mis-executed; the value criterion registers a miss.

Across a vendor. A second weak-tier family, gemma-4-26b in the gemma arm (GM3) under a laboratory-grade instrument, emits the rival argmax itself as its modal output at every discriminating cell. A post-hoc structural diagnostic finds all 27 of 27 rival-valued packings carrying the rival's exact two-radius decomposition, 27 of 43 discriminating-cell validities against the original family's 2 of 57. The cross-vendor subsection and Supplement S2 carry the cells. A family with more arithmetic reach executes the harder in-family construction instead of truncating: same recipe family, other branch.

Not counted as a direction: the top alias. A third nominal tier attempts a more ambitious construction at every cell than either other tier and is valid in 4 of 30 invocations. Its alias-to-weights binding is not attestable and four valid samples cannot carry a tier statement, so it supports nothing here. Appendix D reports the arm, the ladder table and the failure taxonomy in full.

The three directions share the order of failure. Selection is not the bottleneck: the weak tier picks the argmax when shown it, and a family with more reach emits it unprompted. Execution is: the picked construction is mis-built in text, reached but not passed in math-only code, and cleared at rate only where a Sonnet-tier model hands the arithmetic to an optimizer. The template is what survives execution.

5 Zero-shot characterization does not transfer

The weak tier's template is a closed form. Supplement S9 reports the prediction in full: modal at all seven tested N, at four of five held-out N, under a perturbed container and under paraphrase. Four registered arms each changed one thing, a parent, a generation, the serving path or the reasoning budget; two more changed the vendor. The persistence prediction failed under all three changes of setting. The falsifier fired under conditioning; two of four loop predictions were not met, with a third registered open; validity collapsed on the pinned path until the reasoning budget restored it. Every verdict here was registered before the arm's first scored run, post hoc readings excepted; commits, amendments and outcomes are in Table 5, full reports in Supplement S8.

The conditioning arm (arm MU): the anchor dissolves after one step. One-parent mutation calls ran under three parents, the model's own anchor, the in-family rival and an off-family diagonal chain, at $N = 13$, 21 and 31, $n = 15$ per parent and cell, 135 invocations. The falsifier, an anchor rate under the rival parent of at most 20%, scopes the anchoring claim to unconditioned calls. It fired: 0 of 26 valid outputs under the rival parent keep the anchor, and every one keeps or refines the rival. Under the model's own anchor 7 of 28 keep it, against a registered 50% or more. Only the off-family parent is abandoned for the template, in 20 of 34. Post hoc, none of the 63 outputs valid at 10^{-9} clears the family argmax; Supplement S1 has the pool.

The loop arm (arm L): iteration moves the anchor, not past the family. Generation 0 is the bare template; generations 1 to 5 use the conditioning arm's mutation template. Two cells, $N = 13$ and 31, by two archives, GREEDY and DIVERSE, give four lineages of six rounds at population 5, 120 invocations. Four predictions: modal at generation 0 in every lineage; dissolution survives iteration; the greedy archive returns toward the family; and, left open, best-of-run against the family argmax. The second held. The first and third were not met: on-prediction falls from 4 of 9 generation-0 proposals to 5 of 20 conditioned ones at $N = 13$, and from 7 of 8 to 6 of 29 at $N = 31$. The first fails on one lineage under the registered tie rule; the third fails with the direction reversed. No lineage clears the family: zero of 49 valid conditioned outputs exceed the best value the recipe family can express.

The pinned-path arm (arm P): the registered prompt on a bare serving path. This arm sends the registered prompts, byte-identical, through a path that pins temperature: OpenRouter, temperature 1.0, no system prompt. It ran $n = 15$ per cell at the seven square, five held-out and three code cells, 225 invocations, with four predictions and one falsifier. Validity collapsed: 0 of 105 outputs at the square cells are valid at either tolerance, and 4 of 75 at the held-out cells, none clearing. The direct-emission predictions are unscoreable under the arm's floor and the falsifier's denominator is empty. On the code cells 12 of 45 programs are valid and none clears, so the code-channel prediction holds. Two post hoc diagnostics: temperature 0.0, 0.5 and 1.0 gives 0 of 6 valid at each. The same prompt through the agent runtime, same day, gives 6 of 6 valid and lands the anchor $T(4, 13) = 1.625$ in 4 of 6. The thinking-budget arm (arm P-D) names the component with the same $N = 13$ prompt on the pinned path, $n = 15$ each, under D1, extended thinking on, and D2, one system line asking for care. The prediction: D1 valid in at least 8 of 15, D2 in at most 3. D1 is valid in 12 of 14 and D2 in 0 of 13, so it holds, and the anchor $T(4, 13) = 1.625$ is modal among D1's valid outputs at 5 of 12. The anchoring result is a property of its instrument, and the component is the reasoning budget, a request parameter on the pinned path.

The second-vendor arms: a boundary that runs both ways. Two registered arms send the same prompts across vendors; Supplement S2 has them in full. The gemma arm (arm GM3), gemma-4-26b on its first-party API, met its pooled bar: 46 of 80 valid packings, 57.5%, sit on the rule's value against a registered 30%. The pass rides on the non-discriminating cells: at the three discriminating cells the rate is 11 of 43, 25.6%, and the cell-level prediction did not hold. The misses are the rival argmax itself, 27 of 43 discriminating-cell validities carrying its two-radius decomposition, as the mechanism section uses. The cross-vendor arm (arm V) addressed thirteen free-tier aliases, and ten fell below the registered validity floor. Of the three above it, two transfer at point estimate, 8 of 12 and 11 of 18 on-prediction with Wilson intervals spanning the 50% bar; gemma-4-31b does not, at 0 of 15. At the discriminating cells the decomposition gives 3 of 4 and 5 of 9, neither separating from the template-shape null of Supplement S9.2, with exact upper tails 0.111 and 0.145. So the anchor is neither a single-vendor artifact nor universal. With the conditioning, loop and pinned-path arms the boundary is closed from every side the study reached: one tier, one channel, one instrument, one conditioning regime.

6 Related work

LLM-driven discovery and the circle-packing scoreboard. Language Model Crossover (Meyerson et al., 2024), ELM (Lehman et al., 2022), EvoPrompt (Guo et al., 2024), EoH (Liu et al., 2024) and LLaMEA (van Stein & Bäck, 2025) established the LLM call as an EC operator over prompts and programs. FunSearch turned the pattern into a discovery claim, and every system in the introduction's scoreboard inherits it. HELIX's figure is marginally *below* ShinkaEvolve's relaxed-tolerance figure and above its strict one, yet is described as state of the art. ThetaEvolve claims new best-known bounds with a printed $n = 26$ figure equal to HELIX's. The benchmark is saturated across the AlphaEvolve–ShinkaEvolve–HELIX cluster, and the claims outrun the digits. OpenEvolve (Sharma, 2025), the open-source AlphaEvolve reproduction most practitioners run, reports 2.634292 in its published circle-packing example at the time of access; the repository is unpinned. Supplement S9.1 records the correction to our record comparison and its $N > 40$ coverage gap. Two critiques bear on our framing and are often conflated, so we attribute them separately: Gideoni et al. (2026) show simple baselines recover much of the reported advantage, and Berthold et al. (2026) show classical solvers do too.

Template convergence and diversity collapse. "Mutation Without Variation" (Gurkan et al., 2026) finds iterated LLM program mutation collapsing onto previously seen structural templates: in 87% of chains, over 93% of mutations revisit a seen form. Same bias family, different regime: mutation loops, program space, no closed form, no preregistration. It is the closest published neighbour to the loop arm (L). That arm's second registered prediction confirmed the analogous statement inside a scored loop: iterating the archive moves the anchor, and no valid output clears the family argmax. Their result is about structural revisiting; ours is about a value bound the revisited structures cannot cross. On anchoring, Huang et al. (2026) treats it as a general bias over initial information, Lou & Sun (2024) document numeric primes dragging point estimates, and Malberg et al. (2025) test it on a numeric prime among thirty biases. Our modality differs: a construction template has a computable value where a scalar anchor has only a measurable pull. "Measuring the Gap Between Human and LLM Research Ideas" (Chen et al., 2026) reports the same collapse one level up, in idea space. Against 11,683 human ideas, LLM ideas concentrate on bridge-type opportunities, 47–64% versus 12% for humans, with normalized entropy down to 0.55 versus over 0.92. "Artificial Hivemind" (Jiang et al., 2025) documents the same homogeneity across open-ended domains at survey scale. Those results are distributional; ours is the limiting case in which the collapsed mode is a single closed-form-computable object, stated before sampling. Converging skepticism about what the proposer contributes comes from "Dictionaries, Not Darwin" (Li, 2026), Wang et al. (2026), BehaveSim (Zhang & Lu, 2026), Strategy Diversity (Yang et al., 2026) and the bin-packing critiques (Herrmann & Pallez, 2026; Sim et al., 2025). Two results cut the other way. Zhang et al. (2024) grounds the *importance* of evolutionary search empirically, evidence against our substitution of an unconditioned call for the loop. Zhang et al. (2026b) finds strong LLM optimizers act as *local refiners*, which a parent-conditioned call would be. Both are engaged in the limitations section.

Scaling inversions. Zhou et al. (2024) show larger and more instructable models becoming less reliable. The o3/o4-mini system card (OpenAI, 2025) reports a higher PersonQA hallucination rate for o3 than for o1 alongside higher accuracy, attributed to more claims made. Shojaee et al. (2025) reports collapse past a complexity threshold. GeoBuildBench (Kim et al., 2026) finds geometric construction a regime where nominal capability and executed correctness diverge, and MathConstruct (Balunović et al., 2025) benchmarks constructive proofs, where the object built is checkable. Our instantiation is *attractor family* versus executability.

Preregistration lineage. Thomas et al. (2026), Williams et al. (2026), Vaccaro (2026) and Zhang (2026) preregister recipes, outcome-blinded predictions and agent protocols. HindsightBench (Jia, 2026) releases frozen preregistrations of directional aggregate hypotheses. We adopt the standard rather than claim it; what we add is the object locked, exact closed-form point predictions of specific output values. Structural precedents: Kaplan et al. (2020) and Hoffmann et al. (2022) publish a functional form then run the confirming instance; Snell et al. (2024) is the closest structural twin.

7 Limitations and untested alternatives

Scale. The related work's evidence that evolutionary search contributes (Zhang et al., 2024) and that strong LLM optimizers act as local refiners (Zhang et al., 2026b) is answered by two arms. The conditioning arm (MU) takes one conditioning step, and the loop arm (L) runs five generations of a registered archive-and-mutate loop. What stays open is scale: population five over five generations at one tier is a loop, not the loops of the related work. The mechanism-free reading, that the $k \times k$ grid is what survives mental arithmetic, was probed by the argmax-in-hand arm (CH) and split in two. Tractability does not explain the selection; on that arm's post-hoc attempt-level reading it does explain the instantiation.

Contamination is probed at one level only. $N = 26$ is the canonical published cell and plausibly in training data. The held-out-cells, container and recall arms (CN, CP and RP) bound the recall mechanisms; their rows are Supplement S9.4 and Supplement S9.5. They test held-out N, a container that is not the benchmark's, and a direct request for the published values. A planted-canary test is inapplicable to a benchmark we did not author. Absence of the fitted cells from the pretraining corpus is established by none of it.

Vendor scope. All tiers in the main study come from one provider; the cross-vendor subsection moves the pooled-level claim across vendors. One finding is itself a limitation: a cross-vendor null under a fixed output budget is not evidence about the model until the budget's interaction with the vendor's response structure is checked. The 4k-budget gemma arm (GM2) returned 0 of 140 parseable at a 4,096-token budget. The same model at 16,384 tokens parses at 123 of 140, because the vendor API returns deliberation on a separate channel that the smaller budget died inside. After the cross-vendor subsection the anchoring law is cross-vendor at the pooled level and does not separate from Supplement S9.2's template-shape null at the discriminating cells. Its full cell-level mode structure is confirmed only within the original family.

Two instruments, and the contrast that crossed them. The agent runtime does not expose decoding parameters, so every runtime row is a distributional claim over an observed sampling regime. The pinned rows bound that: the pinned-path arm (P), the library arm (CL), the isolating arm (CCP) and the weak top-up (CL-W). The pinned-path arm shows the two paths are not interchangeable for the weak tier, so Table 1 never pools across instruments. The Sonnet contrast crossed them: the Sonnet math-only arm (CCS), 0 of 37, ran on the runtime at 10 seconds, and the library arm, 23 of 25, on the pinned path at 120 seconds. The weak math-only arms (CC, CC2) share the 10-second budget. The 12× gap was registered on purpose, since a 10-second gate measures the timeout rather than the search, which the Sonnet tier's programs can nonetheless be bound by. The isolating arm removes the cross: the weak math-only arms' prompt on the pinned path at 120 seconds returns 0 of 35 clear, none at the argmax. The Sonnet gap is therefore attributed to the library alone. Within the weak tier the contrast is same-instrument, 0 of 12 and 0 of 11, both pinned, but crosses the budget: the pinned-path arm's code cells ran at 10 seconds. No weak-tier program in any arm has passed the argmax under the registered parser at either budget; the lenient reading of the library arms finds 4 of 41.

`opus_alias` provenance. The label denotes the serving alias addressed, not an identified model version. Without pinned weights we cannot separate a model-tier effect from a serving-path effect, and the supporting anomalies live only in the runtime transcript.

Prompt robustness. Every arm other than the paraphrase arm (PP) conditions on a single hash-locked prompt; locking buys reproducibility, not robustness to rewording. The paraphrase arm samples three semantic-preserving paraphrases at the two discriminating cells, and the registered prediction stays modal at all six paraphrase-cells, Supplement S9.5. So the anchor is not keyed to the literal template string at those cells.

Untested alternatives.

- Typicality bias in preference-tuned models (Zhang et al., 2026a) predicts concentration on the most typical construction without reference to geometry; no observation here excludes it.
- Membership and extraction probes of the kind used to detect benchmark contamination (Carlini et al., 2021; Golchin & Surdeanu, 2024) were not run.
- A formatspread-style sweep on the scale of Sclar et al. (2024) remains untested.
- Alignment-induced diversity loss (Kirk et al., 2024), as the account of the cross-vendor concentration differences, which are measured but not explained.

8 Reproducibility, deviations, and disclosures

Provenance in brief; Appendix E carries it in full.

- **Registration and provenance.** Every arm's prompts were hashed and its predictions and falsifiers fixed before its sampling, with the exceptions the deviation table records. No arm was selected on its result, and every arm is reported. The provenance appendix lists each registration with its commit, ledgers and scripts, the limits on what the prompt digests lock, a metadata correction to the shipped ledger, the excluded records, and the internal adversarial check. Deviations are tabled in Appendix C. The post-hoc tests in the body carry no multiplicity correction; their p-values are descriptive, not confirmatory.
- **Use of AI systems.** Claude models were used to draft manuscript prose and to assist implementation. The human author designed the study, set the research questions, specified the instruments and their decision rules, and approved each preregistration before sampling. The author made the final inclusion, stopping and submission decisions and is solely responsible for the content. The claim a reader can check is the ordering, not the authorship: each preregistration commit is a git ancestor of the sampling it governs. The drafting models share a family with arms under study, the Sonnet direct arm among them, so what a reader is asked to trust is the released artifact, not the author.
- **Artifact availability.** The complete artifact accompanies this submission as anonymized supplementary material: ledgers, analysis and reproduction scripts, LP oracle, every preregistration and amendment, frozen reports and the corrections ledger, with the supplement PDF. Every number regenerates via `HOW_TO_RUN.md` from the shipped ledgers. Only the pinned-path arms, P, CL, CCP, CL-W and P-D, can be re-sampled. The runtime rows depend on a dispatch wrapper and inherited context the digests do not lock, as the registration protocol states, so a reader can re-score them but cannot re-draw them. The working repository is a public remote, unlinked here for anonymity.
- **Which arms are in the body.** Every registered arm has a row in Table 4 and a ledger named in the section that reports it or in the provenance appendix. An arm has a body subsection when its registered verdict bears on the channel table, the mechanism or the transfer boundaries. Arms whose verdicts rest on non-discriminating cells, on fewer valid samples than their floor, or on a null that does not separate from a template-shape null are reported outside the body. The trace and rectangle arms (T and G) are supplement-only, in Supplement S6 and Supplement S5; the optimizer-alone baseline (B) has its reference program listed verbatim in Supplement S7. The cross-vendor and gemma arms (V and GM3) are split. Their registered pooled verdicts rest on non-discriminating or below-floor cells and are reported in full in Supplement S2; their discriminating-cell findings carry a paragraph in the cross-vendor subsection. The extend-cell arm (M) keeps its registered verdicts and its per-cell decomposition in Supplement S4. The `opus_alias` post-hoc dual accounting is Supplement S3. The

pooled-zero table of Supplement S1 pools body arms and is not an arm. The criterion moved no number.

- **Stopping rule.** The manuscript closed after implementing the internal-check findings; no further data-dependent analysis was performed. The pinned-path arm (P) and the library arm (CL) were registered together after an external review and pushed before either was sampled; each was sampled and scored once. The pinned-path arm's two post-hoc diagnostics are labelled as such in Table 5. After that close, the arms below were each registered, run and scored once under their own preregistrations, each analysis script committed before its sampling and not modified after. They are the weak math-only arms (CC, CC2), the Sonnet math-only arm (CCS), the held-out-cells arm (CN), the container, recall and paraphrase arms (CP, RP and PP) and the loop arm (L). The recall arm ran with its control amendment. The isolating arm (CCP), the weak top-up (CL-W) and the thinking-budget arm (P-D) were each registered in response to an outside read and pushed before their first calls. The optimizer-alone baseline (B) was registered the same way, run under one amendment recorded in the deviation table, and scored once. The analyses named but not run are recorded in Supplement S6.5 and Supplement S9.3.

References

Mislav Balunović, Jasper Dekoninck, Nikola Jovanović, Ivo Petrov, and Martin Vechev. Mathconstruct: Challenging llm reasoning with constructive proofs, 2025. URL `https://arxiv.org/abs/2502.10197`.

Timo Berthold, Dominik Kamp, Gioni Mexi, Sebastian Pokutta, and Imre Polik. Out-of-the-box global optimization for packing problems: New models and improved solutions, 2026. URL `https://arxiv.org/abs/2605.04850`.

Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *30th USENIX Security Symposium*, 2021.

Mert Cemri, Shubham Agrawal, Akshat Gupta, Shu Liu, Audrey Cheng, Qiuyang Mang, Ashwin Naren, Lutfi Eren Erdogan, Koushik Sen, Matei Zaharia, Alex Dimakis, and Ion Stoica. Adaevolve: Adaptive llm driven zeroth-order optimization, 2026. URL `https://arxiv.org/abs/2602.20133`.

Ziyu Chen, Yilun Zhao, and Arman Cohan. Measuring the gap between human and llm research ideas, 2026. URL `https://arxiv.org/abs/2607.01233`.

Yonatan Gideoni, Sebastian Risi, and Yarin Gal. Simple baselines are competitive with code evolution, 2026. URL `https://arxiv.org/abs/2602.16805`.

Shahriar Golchin and Mihai Surdeanu. Time travel in LLMs: Tracing data contamination in large language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2308.08493.

Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.08532.

Can Gurkan, Forrest Stonedahl, and Uri Wilensky. Mutation without variation: Convergence dynamics in llm-driven program evolution, 2026. URL `https://arxiv.org/abs/2606.05408`.

Julien Herrmann and Guillaume Pallez. An in-depth study of llm contributions to the bin packing problem, 2026. URL `https://arxiv.org/abs/2510.27353`.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models, 2022. URL `https://arxiv.org/abs/2203.15556`.

Yiming Huang, Biquan Bie, Zuqiu Na, Weilin Ruan, Songxin Lei, Yutao Yue, and Xinlei He. Understanding the anchoring effect of llm with synthetic data: Existence, mechanism, and potential mitigations, 2026. URL https://arxiv.org/abs/2505.15392.

Haozhe Jia. Hindsightbench: A black-box behavioral audit protocol for parametric hindsight in time-indexed llm decision tasks, 2026. URL https://arxiv.org/abs/2607.18867.

Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond), 2025. URL https://arxiv.org/abs/2510.22954.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. URL https://arxiv.org/abs/2001.08361.

Valentin Khrulkov, Andrey Galichin, Denis Bashkirov, Dmitry Vinichenko, Oleg Travkin, Roman Alferov, Andrey Kuznetsov, and Ivan Oseledets. Gigaevo: An open source optimization framework powered by llms and evolution algorithms, 2025. URL https://arxiv.org/abs/2511.17592.

Jinwoong Kim, Rui Yang, and Huishuai Zhang. Geobuildbench: A benchmark for interactive and executable geometry construction from natural language, 2026. URL https://arxiv.org/abs/2605.13167.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.06452.

Robert Tjarko Lange, Yuki Imajuku, and Edoardo Cetin. Shinkaevolve: Towards open-ended and sample-efficient program evolution, 2025. URL https://arxiv.org/abs/2509.19349.

Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.

Joel Lehman, Jonathan Gordon, Shawn Jain, Kamal Ndousse, Cathy Yeh, and Kenneth O. Stanley. Evolution through large models, 2022. URL https://arxiv.org/abs/2206.08896.

Pan Li. Dictionaries, not darwin: Set-level selection beats llm evolution in scientific equation discovery, 2026. URL https://arxiv.org/abs/2607.04108.

Fei Liu, Xialiang Tong, Mingxuan Yuan, Xi Lin, Fu Luo, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. Evolution of heuristics: Towards efficient automatic algorithm design using large language model. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2401.02051.

Jiaxu Lou and Yifan Sun. Anchoring bias in large language models: An experimental study, 2024. URL https://arxiv.org/abs/2412.06593.

Simon Malberg, Roman Poletukhin, Carolin M. Schuster, and Georg Groh. A comprehensive evaluation of cognitive biases in llms, 2025. URL https://arxiv.org/abs/2410.15413.

Elliot Meyerson, Mark J. Nelson, Herbie Bradley, Adam Gaier, Arash Moradi, Amy K. Hoover, and Joel Lehman. Language model crossover: Variation through few-shot prompting, 2024. URL https://arxiv.org/abs/2302.12170.

Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.

Alexander Novikov, Ngân Vũ, Marvin Eisenberger, et al. Alphaevolve: A coding agent for scientific and algorithmic discovery, 2025. URL https://arxiv.org/abs/2506.13131.

OpenAI. OpenAI o3 and o4-mini system card. https://openai.com/index/o3-o4-mini-system-card/, 2025.

Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024. doi: 10.1038/s41586-023-06924-6.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.11324.

Asankhaya Sharma. Openevolve: an open-source implementation of AlphaEvolve. GitHub repository, `https://github.com/codelion/openevolve`, 2025. Repository unpinned; circle-packing example value read at time of access, 2026.

Parshin Shojaee, Iman Mirzadeh, Keivan Alizadeh, Maxwell Horton, Samy Bengio, and Mehrdad Farajtabar. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity, 2025. URL `https://arxiv.org/abs/2506.06941`.

Kevin Sim, Quentin Renau, and Emma Hart. Beyond the hype: Benchmarking llm-evolved heuristics for bin packing, 2025. URL `https://arxiv.org/abs/2501.11411`.

Charlie Snell, Eric Wallace, Dan Klein, and Sergey Levine. Predicting emergent capabilities by finetuning, 2024. URL `https://arxiv.org/abs/2411.16035`.

Chang Su, Zhongkai Hao, Zhizhou Zhang, Zeyu Xia, Youjia Wu, Hang Su, and Jun Zhu. Helix: Evolutionary reinforcement learning for open-ended scientific problem solving, 2026. URL `https://arxiv.org/abs/2603.07642`.

Maria Thomas, Kristina Gligoric, and Nihar B. Shah. Mitigating llm-based p-hacking by preregistering for the next llm, 2026. URL `https://arxiv.org/abs/2606.27687`.

Michelle Vaccaro. Preregistration for experiments with ai agents, 2026. URL `https://arxiv.org/abs/2606.11217`.

Niki van Stein and Thomas Bäck. Llamea: A large language model evolutionary algorithm for automatically generating metaheuristics, 2025. URL `https://arxiv.org/abs/2405.20132`.

Haorui Wang, Parshin Shojaee, Kazem Meidani, Kunyang Sun, José Miguel Hernández-Lobato, Teresa Head-Gordon, Jiajun He, Chandan K. Reddy, Chao Zhang, and Yuanqi Du. Towards diverse scientific hypothesis search with large language models, 2026. URL `https://arxiv.org/abs/2606.10587`.

Yiping Wang, Shao-Rong Su, Zhiyuan Zeng, Eva Xu, Liliang Ren, Xinyu Yang, Zeyi Huang, Xuehai He, Luyao Ma, Baolin Peng, Hao Cheng, Pengcheng He, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. Thetaevolve: Test-time learning on open problems, 2025. URL `https://arxiv.org/abs/2511.23473`.

Marcus Williams, Hannah Sheahan, Cameron Raymond, Tomek Korbak, Deng Pan, Peilin Yang, Leon Maksin, Ningyi Xie, Phillip Guo, Ian Kivlichan, and Micah Carroll. Predicting llm safety before release by simulating deployment, 2026. URL `https://arxiv.org/abs/2607.07184`.

Xia Yang, Xuanyi Zhang, Hao Hu, and Feng Ji. Beyond accuracy: Evaluating strategy diversity in llm mathematical reasoning, 2026. URL `https://arxiv.org/abs/2605.09292`.

Dong Zhang. Testing frontier large language models' physics literacy in parallel physical worlds, 2026. URL `https://arxiv.org/abs/2607.00276`.

Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock llm diversity, 2026a. URL `https://arxiv.org/abs/2510.01171`.

Rui Zhang and Zhichao Lu. Rethinking code similarity for automated algorithm design with llms, 2026. URL `https://arxiv.org/abs/2603.02787`.

Rui Zhang, Fei Liu, Xi Lin, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. Understanding the importance of evolutionary search in automated heuristic design with large language models, 2024. URL `https://arxiv.org/abs/2407.10873`.

Xinhao Zhang, Xi Chen, François Portet, and Maxime Peyrard. What makes an llm a good optimizer? a trajectory analysis of llm-guided evolutionary search, 2026b. URL `https://arxiv.org/abs/2604.19440`.

Lexin Zhou, Wout Schellaert, Fernando Martínez-Plumed, Yael Moros-Daval, Cèsar Ferri, and José Hernández-Orallo. Larger and more instructable language models become less reliable. *Nature*, 634(8032):61–68, 2024. doi: 10.1038/s41586-024-07930-y.

A Prompts, verbatim

A.1 Bare template (`{n}` parameterized; the $N = 13$ instance hashes to `32db485bea625ff9f39f4723ebf1a0 1f337559a9e2cf567fb486928f71f7f8df`):

```
Pack exactly {n} non-overlapping circles inside the unit square [0,1]x[0,1]
so that the sum of their radii is as large as possible.
MUST hold: exactly {n} circles; every circle fully inside the unit square
[0,1]x[0,1] (x-r >= 0, x+r <= 1, same for y); no two circles overlap
(distance between centers >= sum of the two radii; touching is allowed).
Do not write or execute code - construct the packing by reasoning alone.
Output ONLY the raw Python list of {n} [x, y, r] lists. No explanation, no
code fences, no other text.
```

A.2 `trace_v2` template — identical to A.1 through line 3, then:

```
First write one line beginning "METHOD:" naming the construction you used.
Do not use square brackets anywhere in that line.
After the METHOD line, output ONLY the raw Python list of {n} [x, y, r]
lists. No explanation, no code fences, no other text after the list.
```

A.3 Dispatch wrapper (both arms; *not* part of the hashed task prompt, hence an instance of the fragment-hashing limitation in §3.1): `Do not use any tools.  Your entire final message must be the answer and nothing else.`

A.4 Registered prompt hashes. bare N=13 `32db485b...`, N=17 `8437df75...`, N=21 `a415425b...`, N=31 `a664d003...`, N=35 `3b08c56e...`, N=37 `2c0a8819...`; trace_v2 N=13 `a920f1c9...`, N=21 `91205727...`, N=31 `bd490b7b...`. Arm CN (held-out N): N=50 `0b858558...`, N=58 `200c0bb4...`, N=62 `b9854a84...`, N=65 `5fb5fefd...`, N=75 `a5db8fc3...`, registered at commit `69a1642`. The N=43 bare hash `1208e7d2...` was ledger-only until arm P registered it (§5), and the pilot prompt was never hashed (§8).

A.5 Parser. Completions are parsed with `ast.literal_eval` after stripping code fences; a completion failing to yield a list of N numeric triples is recorded `literal_eval_failed` and scored invalid, with the failure mode logged. Radii must be strictly positive; containment and pairwise non-overlap are checked at the two registered tolerances.

B The registered arms

Every registered arm, with the section that reports it, its proposer, design, registered test and outcome. Table 2 in §3 is the subset of these rows that carries Table 1.

Table 4: The twenty-six registered arms: section (a letter is an appendix; Sn is a section of the supplement), proposer, design, registered test, outcome.

Arm	§	Proposer(s)	Design	Registered test	Outcome
F (square, bare)	3.3, S9	weak tier	7 N × 5–20	P1–P5 exact-value modes	modal at 7/7 N
S (Sonnet direct arm)	3.3	Sonnet tier	3 N × 10	P-S1–P-S4 tier contrasts	1/30 on-prediction, 100%/90% valid ($10^{-6}/10^{-9}$); one out-of-family escape; 5 of 20 samples at $N = 13/21$ seen before registration, disclosed
O (top alias)	D, S3	`opus_alias`	3 N × 10	P-O1–P-O4	13% valid; 3 predictions not evaluable (deviation, Table 5)
T (trace)	S6	weak tier	3 N × 20 × 2 arms	P-T1–P-T4 + falsifier	falsifier **triggered** (tie reading); P-T3 fragile
G (rectangle)	S5	16 invocations	2 cells	out-of-sample transfer	partial, 5/11; formula negative result kept
M (extend-cell arm)	3.3, S4	weak tier	4 N × 15	P-M1–P-M4 + F-M1/F-M2	F-M1 **triggered 3/3**: extend branch dies at $m \geq 4$; fifth k confirms; two out-of-family escapes at $N = 20$ (post hoc)
MU (mutation)	5, S8	weak tier	3 parents × 3 N × 15	P-MU1–P-MU4 + F-MU1	F-MU1 **triggered**: anchor dissolves under conditioning
CH (choice)	4.1	weak tier	3 N × 15	P-CH1 vs P-CH2	P-CH1 holds 2/3, P-CH2 1/3 (registered); post-hoc attempt reading: 36/45 reach for the argmax, 30/30 evaluable invalid rows at its order
GM (gemini-flash-lite)	7	2nd vendor	7 N × 20	GM-chain bars	UNDERPOWERED (20 valid / 140)
GM2 (gemma, 4k budget)	7	2nd vendor	7 N × 20	GM-chain bars	0/140 parseable — budget confound, diagnosed by GM3
GM3 (gemma, 16k budget)	5, S2, S8	gemma-4-26b	7 N × 20	P-GM3.1–.3 + falsifier	pooled bar met (57.5%) but discriminating cells $11/43 = 26\%$, below it; misses are the rival construction
V (cross-vendor zero-shot)	5, S2, S8	13 free-tier aliases	5 N × 5	floor + V1/V2	2 vendors TRANSFER, gemma-4-31b DOES-NOT-TRANSFER, V2 HOLDS
CC (weak math-only arm)	3.4	weak tier	3 N × 15	P-CC1 vs P-CC2	P-CC1 holds: anchor stays modal; 0/37 above the family argmax
CC2 (replication arm)	3.4	weak tier	3 N × 15	R1 + R2 + F-CC2.1	**REPLICATED**: 0/41 above argmax, anchor modal 3/3
CCS (Sonnet math-only arm)	3.4	Sonnet tier	3 N × 15	P-CCS1 vs P-CCS2	P-CCS2 holds: 0/37 above argmax — the channel, not the tier
CN (held-out N)	4.2, S9.4	weak tier	5 N × 15	P-CN1 vs P-CN2 + S-CN1–3	P-CN1 holds: mode at 4/5 held-out cells, 3/3 discriminating; structure 5/5
CP (perturbed container)	S9.5	weak tier	5 N × 15	P-CP1 vs P-CP2 + S/F	P-CP1 holds: mapped prediction modal 4/5; S-CP1 not met (4/61 rival)
RP (direct recall)	S9.5	weak tier	6 N × 15	P-RP1 vs P-RP2 + F-RP1	P-RP1 holds: 0/44 recalls; 87/87 scored responses UNKNOWN

Arm	§	Proposer(s)	Design	Registered test	Outcome
PP (paraphrase)	S9.5	weak tier	3 rewrites × 2 N × 15	P-PP1 vs P-PP2 + S-PP1/2	P-PP1 holds 6/6: modal under every paraphrase; S-PP1 not met (6/75 rival)
L (iterated loop)	5, S8	weak tier	2 N × 2 archives × 6 rounds × 5	L1–L4 + F-L1	L2 holds; L1 and L3 **not met** (L3 reversed); L4 open: 0/49 conditioned outputs clear the family
P (pinned path)	5, S8	weak tier, bare API	15 cells × 15	P-P1–P-P4 + F-P1	validity collapses: 0/105 valid at the square cells, 4/75 held-out; every direct-emission prediction UNSCOREABLE; P-P3 holds (0 of 12 valid programs clear); same-day agent-runtime control 6/6 valid, anchor 4/6 (post hoc)
CL (library code)	3.5	weak + Sonnet, numpy/scipy	2 tiers × 3 N × 15	P-CL1 vs P-CL2 + F-CL1	weak: 0/11 clear, P-CL1 holds; Sonnet: **23/25 clear at 3/3 cells**, P-CL2 holds, **F-CL1 fires**
CCP (isolating arm)	3.5	Sonnet tier, bare API, math only, 120 s	3 N × 15	P-CCP1 vs P-CCP2 + F-CCP1	P-CCP1 holds: 0/35 clear, none reaches the argmax; best per cell 1.625, 2.1, 2.673 — the library, not the path or the budget
CL-W (weak top-up)	3.5	weak tier, numpy/scipy	3 N × 15, pooled with CL	P-CLW1 vs P-CLW2 + F-CLW1; lenient reading secondary	P-CLW1 holds: 0/19 pooled clear, upper bound 16.8%; lenient reading 4/41, dead zone
B (optimizer alone)	3.5, S7	no model in the loop; a fixed SLSQP program	3 N × 15 seeds, 120 s	P-B1 vs P-B2 + S-B1	P-B1 holds: **45/45 clear**, 3/3 cells; S-B1: beats the Sonnet best at 3/3 cells; amendment 1 (version 1 blocked by the allowlist, kept)
P-D (runtime component)	5, S8	weak tier, bare API, $N = 13$	2 conditions × 15	P-PD1 vs P-PD2 vs P-PD3 + S-PD1	P-PD1 holds (closed at 27/30 rows by amendment, verdict unchangeable): thinking on 12/14 valid, template modal; system line only 0/13 — the reasoning budget is the component

C Deviations and disclosure table

Table 5: Deviations from preregistration.

Prediction	Registered rule	As registered?	Outcome / reason
P1–P3	exact-value point predictions	yes	Supplement S9.3, Supplement S5
P4 (N=37)	converge, 3.0345178	yes	confirmed
P5 (N=35)	truncate, 2.9166667, structural separation	yes	3/4 valid; $n = 4$, "consistent with"

continued on next page

Prediction	Registered rule	As registered?	Outcome / reason
P-S1. . . P-S4 (arm S)	directional tier comparisons	yes; the registration (`arm_s_preregistration.txt`) disclosed that 5 of 20 samples at $N = 13/21$ had been seen before registration and that $N = 31$ was fully blind	valid 30/30 at 10^{-6} and 27/30 at 10^{-9}; on-prediction 0/10, 0/10, 1/10; one out-of-family escape at $N = 31$ (§3.3)
P-O1, P-O2, P-O4	tier comparisons on on-prediction / rival rates	**no**	declared not evaluable post hoc after a 13% validity collapse; the registered disconfirmation (regression toward the trap) did not occur
P-O3	multi-radius fraction ≥ 0.9 of valid	yes	trivially satisfied on 4/4 valid
P-T1	direction at each N **and** pooled $p < 0.05$	yes	not confirmed, $p = 0.30$
P-T2	directional, no alpha	yes	confirmed as registered (1/53 vs 2/50); no inferential weight claimed
P-T3	directional, no alpha	yes	met as registered; $p = 0.0325$ uncorrected fails Holm; carried by $N = 13$ (Supplement S6.4)
P-T4	$\geq 90\%$ of scoreable claims match	yes, plus corrected scorer	38/41 (93%) original; 54/56 (96.4%) blind, frozen rubric
P-M1–P-M3	modal output $= V(k^*, m)$ at $N = 20/30/41$	yes	**disconfirmed at all three cells** — modal outputs are $(k^*{+}1)$-grid values (Supplement S4)
P-M4	$N = 57$ modal $= T(8, 57)$, rival count 0	yes	confirmed (6/10 on-prediction, 0/10 rival)
F-M1 (arm M falsifier, filler branch)	modal $\neq V(k^*, m)$ at 2+ of 3 converge cells, ties count as NOT	yes, tie-stated in registration	**triggered at 3 of 3**; $V(k, m)$ restricted to $m \leq 1$ (Supplement S4)
F-M2 (arm M falsifier, truncate branch)	$N = 57$ modal $\neq T(8, 57)$	yes	not triggered
Arm M rejection rule	runtime deaths excluded, counted	yes (registered for this arm)	3 $N = 41$ invocations excluded, cell reported as 12 of 15
GM3 serving path	first-party API only	amendment registered **before resumption** (`arm_gm3_amendment1_openrouter.md`, `67cc4a1`)	41/140 rows via routed path, provider logged; 59.0% / 57.5% dual accounting, no verdict change (§5)
Arm V proposer set	3 opencode aliases	expanded 3 → 13 by amendments 1–2, each registered **before the sampling it governs**	verdicts in §5; `big-pickle` disclosed as vendor-unattested in amendment 1
Arm V transport repair	single pass, `max_tokens` 8192	amendment 3 registered **before resumption** (mechanical: 24,576 budget, retry, censored-row accounting)	877 attempt rows over the 325 registered slots (314 carry rows); floor evaluated per slot (§5)
GM3 rival-structure check	—	**post hoc**, disclosed (`diagnostics_gm3_rival.py`)	27/27 rival-valued discriminating-cell packings carry the rival's two-radius structure (§5); the registered signature is scoped to k^* and cannot see $(k^*{-}1)$ structure
Falsifier	trace_v2 validity `<=` bare at 2+ of 3 N	**both readings reported**	**triggered** under the registered tie-inclusive reading (2/3 cells); not triggered under the strict `<` the code implemented (Supplement S6.3)
Ladder inclusion	—	**post hoc**	`opus_alias` initially excluded, later included after results known; ladder reported both ways (Supplement S3)
Cap exclusion	—	**not registered**	five records excluded; both rates reported (Supplement S9.3)
Arm CN ledger transcription	verbatim rows	**disclosed** (`arm_cn_results.txt`)	eleven $N = 65$ rows reconstructed from the filler triple, flagged per row; discarded, P-CN1 reads 3 of 4 (Supplement S9.4)
Arm M out-of-family sweep	—	**post hoc**, not preregistered; found by a sweep disclosed as such	three converge-cell rows ($N = 20$ twice, $N = 30$) above the family best by 1.5–2.0 $\times 10^{-2}$; not modal; the two $N = 20$ rows are valid at 10^{-9} and are the arm-M escapes Table 1 counts, the $N = 30$ row fails the 10^{-9} gate (Supplement S4)

continued on next page

Prediction	Registered rule	As registered?	Outcome / reason
Arms CC2 and CCS (registered follow-ups)	CC2 verdict map fixed in advance; CCS Sonnet-tier probe	registered together **after arm CC's results were known**, disclosed as such; 45 invocations each on the byte-identical prompts	CC2 **REPLICATED** under its pre-stated verdict map: 0 of 41 valid outputs above the rival, anchor bucket modal at 3 of 3 cells; CCS 0 of 37 clear (§3.4)
Arm CC2 tool use	no tools (wrapper A.3)	**disclosed** (`arm_cc2_results.txt`)	one $N = 13$ invocation used tools; scored as emitted (arm-M convention)
Uniform-template null	—	**post hoc**, not preregistered	discriminating-cell rates exceed the 1/3 null; binomial tails 0.043 to 2.9×10^{-4} (Supplement S9.2) (`diagnostics_template_null.py`)
Arm CP (perturbed container)	P-CP1/P-CP2 + S-CP1/S-CP2 + F-CP1	registered **before sampling** (`105e8b7`); drafted at review stage, disclosed	P-CP1 **confirmed** (modal 4/5, $N = 75$ a registered tie against); S-CP1 **not met** (4/61 rival); S-CP2 5/5; F-CP1 not triggered (Supplement S9.5)
Arm RP (direct recall)	P-RP1/P-RP2 + F-RP1	registered **before sampling** (`105e8b7`); draft cell N=32 → N=30, documented	P-RP1 **confirmed**: 0/44 recalls, 87/87 scored responses UNKNOWN; F-RP1 not triggered (Supplement S9.5)
Arm RP positive control	C-RP1 vs C-RP2	amendment 1 registered **before sampling** (`15555eb`), responding to a review finding	C-RP1: 14/15 state 0.5 at N=1; abstention-prior alternative rejected (Supplement S9.5)
Arm PP (paraphrase)	P-PP1/P-PP2 + S-PP1/S-PP2	registered **before sampling** (`2e89614`); paraphrases authored results-aware, disclosed	P-PP1 **confirmed** 6/6 modal; S-PP1 **not met** (6/75 rival); S-PP2 5/6 (Supplement S9.5)

continued on next page

Prediction	Registered rule	As registered?	Outcome / reason
Arm L (iterated loop)	L1–L4	registered **before sampling** (`c9645fe`), scored once (`750bb32`); one pooled bar per cell rather than four independent bars, a design choice that cost the arm power and is not repaired after the fact; one post-sampling correction disclosed: the registered configuration stored the family argmax to seven decimals while the clearance test compares at 10^{-9}, so a proposal landing exactly on the rival at $N = 31$ tested as above the truncated literal by 3.7×10^{-8}; the scorer now recomputes the argmax in closed form, agreeing with the stored value to 1.3×10^{-15}, and the change affects only the 10^{-9} clearance test (corrections ledger item 34)	L2 **confirmed**; L1 **not met** on 13-DIVERSE, a 2–2 tie at $n = 4$ decided against the prediction by the registered tie rule; L3 **not met and reversed**, the prediction withdrawn rather than reframed; L4 open and answered — 0 of 49 valid conditioned outputs clear the family argmax (§5)
Arm P (pinned path)	P-P1–P-P4 + F-P1	registered **before sampling** (`aa473ad`); post-review motivation disclosed	225 invocations, 219 returning content and 6 refused by the serving path and recorded as such; validity collapsed to 0/105 at the square cells; direct-emission predictions **not evaluable** under the registered floor; P-P3 holds (0/12); F-P1 denominator empty, reported as such (§5)
Arm P temperature diagnostic	—	**post hoc**, not preregistered	0/6 valid at temperature 0.0, 0.5 and 1.0; temperature excluded as cause (§5)
Arm P runtime control	—	**post hoc**, not preregistered	same prompt through the agent runtime, same day: 6/6 valid, anchor 4/6 (§5)
Arm P-D (runtime component)	P-PD1/P-PD2/P-PD3 + S-PD1	registered **before sampling** (`5554cc0`), after an outside review asked for a diagnostic beyond temperature; run interrupted at 27 of 30 rows (14 D1, 13 D2) when the serving path refused the last three for credit; closed there by amendment 1 (`arm_pd_amendment1_closed_at_27.md`), since no completion could change the verdict	P-PD1 **holds** and the last three rows could not change it: thinking on 12/14 valid, modal $T(4, 13)$; system line 0/13; S-PD1 met (§5)

continued on next page

Prediction	Registered rule	As registered?	Outcome / reason
Arm B (optimizer alone)	P-B1/P-B2 + S-B1	registered **before the first scored run** (`1932cc0`); post-review motivation disclosed; amendment 1 (`c486843`): version 1 bounded its restarts by wall clock and imported `time` and `sys`, which the registered allowlist forbids, so the AST gate blocked 45/45 and that run is kept in the artifact; version 2 fixes the restart count at 50 and changes nothing else, 50 chosen because an unscored timing run outside the gate fit 80 restarts in 95 seconds at $N = 31$; both versions committed before the single scored run, no result informed either; authorship Claude drafting under the human author (§8)	P-B1 **holds**: 45/45 clear, 3/3 cells at the 20% bar; S-B1: best exceeds the Sonnet best at 3/3 cells; contribution 1 rewritten per the registered rule (§3.5)
Arm CL (library code)	P-CL1/P-CL2 + F-CL1	registered **before sampling** (`aa473ad`); one execution-mechanism correction made and disclosed in the same commit: an earlier draft of the registration passed the libraries to the child through `PYTHONPATH`, which `python -I` ignores, so a fixed driver committed with the analysis script inserts the interpreter's site-packages into `sys.path` and runs the model's source unmodified via `runpy`; the driver is identical for every row, not part of the scored source, and predates any sampling	90 invocations, 4 (all Sonnet-tier) refused by the serving path with an HTTP 402 and recorded as such, 85 parseable programs, 1 Sonnet parse failure; weak tier P-CL1 holds (0/11); Sonnet tier P-CL2 holds, **F-CL1 fired** (23/25, 3/3 cells); registered integration rule applied (§3.5)
Arm CL lenient reparse	—	**post hoc**, not preregistered	the 22 weak-tier parse failures are all `numpy` scalar reprs; stripping the wrapper recovers 9 valid programs (3, 4, 2 by cell) and 1 clearance at $N = 13$ (1.812531608); registered 0 of 11 stands, 1 of 20 reported beside it (§3.5) (`cl_lenient_reparse.py`)

continued on next page

Prediction	Registered rule	As registered?	Outcome / reason
Arm CCP (isolating arm)	P-CCP1/P-CCP2 + F-CCP1	registered **before sampling** (`ebf213e`); post-review motivation disclosed; one 402 interruption at row 24, resumed with one worker, request unchanged (`32fd3c7`)	P-CCP1 **holds**: 0/35 clear, none at the argmax; F-CCP1 not triggered; registered integration rule applied (§3.5)
Arm CL-W (weak-tier top-up)	P-CLW1/P-CLW2 + F-CLW1 + S-CLW1	registered **before sampling** (`f73222a`); registered after the lenient reparse; pooling rule, both readings and the predictions fixed in advance; 45 invocations (15 per cell) rather than the drafted 90, credit-limited, disclosed; the 90 rows of both runs scored in one pass by arm CL's pipeline	P-CLW1 **holds**: 0/19 pooled clear (Wilson upper 16.8%); S-CLW1 4/41 lenient, dead zone; F-CLW1 not triggered (§3.5)
Wave stratification of the bare arm	—	**post hoc**, not preregistered	the trace effect is carried by $N = 13$ and confounded with collection wave; a diagnosis, no confirmatory weight (Supplement S6.4)
Arm MU clearance reread	—	**post hoc**, not preregistered	the clearance question was never registered for arm MU; 88 valid at the primary tolerance, 18 of them above the argmax by at most 1.5×10^{-6} and every one failing the 10^{-9} gate; 63 valid at 10^{-9}, none above the argmax (§5) (`arm_mu_ceiling.py`)
Arm CH attempt-level reading	—	**post hoc**, not preregistered	36/45 reach for the argmax against the bare arm's 5/57; the control comparison over all evaluable rows is 36 of 44 (the radical-literal row is not evaluable and is excluded from both denominators) against 5 of 57, Fisher one-sided $p = 2.2 \times 10^{-14}$ (`arm_f_attempt_uncond.py`); the invalid-row diagnostic, 3 of 7 against 30 of 30, $p = 5.3 \times 10^{-4}$ (`arm_f_attempt_control.py`), conditions on a collider and carries no evidential weight; both are post-hoc strata, and the registered per-cell metric is reported first (§4.1)
Structural k back-out	—	**post hoc**, not preregistered	50/50 on-prediction samples k^*-structured, 64/69 overall (Supplement S9.2) (`diagnostics_kmatch.py`)
Ambition diagnostic	—	**post hoc**, not preregistered	distinct radii and off-lattice fraction rise monotonically across tiers (§4.2) (`diagnostics_ambition.py`)

D The `opus_alias` arm: the ladder and the failure taxonomy

The top-alias arm (O) was valid in 4 of 30 invocations, 13%, at both tolerances. It attempted more ambitious constructions than either other tier and failed geometrically, not numerically. Table 6 is the ladder §4.2 quotes; the two post-hoc decisions behind it and the two corpora behind its first rung are Supplement S3.

The top tier attempts recursive constructions and fails to build them. We name the arm `opus_alias` and make no claim about which weights served it. The runtime accepts only the bare alias and exposes no dated identifier, so the binding is a vendor promise, not an attestable fact. No version number appears in this paper. Completion-time and token-count anomalies were seen in the unreleased session transcript. The ledger has no latency or token field, so they appear in no released artifact and nothing below relies on them.

The registration, `arm_o_preregistration.txt`, was written fully blind. At $N = 13$ the arm built quarter-circle corners at $r = 0.25$ with Apollonius-style fillers in 10 of 10 samples. At $N = 21$ it built mixed-radius grids of roughly 4×4. At $N = 31$ it built a coarse 3×3 grid at $r \approx 1/6$ with border strips, 0 of 10 valid.

Table 6: Three-attractor ladder. Valid columns count samples passing containment and non-overlap at the 10^{-6} and 10^{-9} tolerances; On-pred and Rival count valid samples landing on the predicted value and on the rival argmax. The matched-cell row is the comparison to quote across tiers (83% → 100% → 13%). ‡ Addressed through a bare serving alias; the alias-to-weights binding is not attestable, and that row is not a statement about any released model version. Tier denominators are not matched, the weak tier spanning seven N against three, so the matched-cell row is given separately. Sources: `arm_f_candidates_v2.jsonl` and the three preregistration files.

Tier	Cells	n	Attempted family	Valid 10^{-6}	Valid 10^{-9}	On-pred	Rival	Failure mode
haiku (bare)	13, 17, 21, 31, 35, 37, 43	45	uniform grid, truncated; fillers when underfull	35 (78%)	29 (64%)	12/16 (3 matched cells)	2	overlap 6, outside 1, parse 3
haiku, matched cells	13, 21, 31	60	as above	50 (83%)	—	35/50	2	overlap 5, outside 2, parse 3
sonnet	13, 21, 31	30	perturbed grid, hexagonal rows, 2–3 radii	30 (100%)	27 (90%)	1/30	5 (structure)	none
`opus_alias` ‡	13, 21, 31	30	quarter-circle corners, Apollonius fillers, coarse grid + border strips	4 (13%)	4 (13%)	0/4	0	overlap 24, non-positive radius 2

Failures are geometric: edge strips at $r = 0.03$ sat 0.138 from an $r = 1/6$ grid circle that needed 0.197, and twice the arm padded to count with zero-radius circles. Valid samples score *below* the trap they were expected to fall into, 1.26 to 1.41 at $N = 13$ against 1.625. A post-hoc diagnostic, labelled as such in the repository, finds no tolerance artifact behind the 13%. 24 of 26 invalid samples overlap grossly: median maximum overlap 3.3×10^{-2}, and radius-shrink repair costs a median 15% of the sum of radii. Tolerance-scale near-misses, under 2.5×10^{-5}, fall instead in 5 of 7 geometry-scored weak-tier failures: the exact-tangency grids, not the ambitious constructions.

E Provenance, in full

Section 8 summarises this appendix; here every registration, commit, ledger and script is in one place.

- **Registration.** Prompts were hashed with SHA-256 and the hashes recorded before sampling, with the predicted values to be tested. Bare square arm (F): the five exact-value predictions in the header of `arm_f_repro.py`. Sonnet direct arm (S) and top-alias arm (O): `arm_s_preregistration.txt` and `arm_o_preregistration.txt`. Trace arm (T): `arm_t_preregistration.txt`, hash `ab7900a8...`. Extend-cell arm (M): `arm_m_preregistration.txt` with `arm_m_prompts.json`, committed at `e528c6b` before sampling. Conditioning arm (MU): `arm_mu_preregistration.txt` with `arm_mu_prompts.json`, committed at `2b7d202` before sampling; decision thresholds, power analysis, tie conventions and falsifier all registered. Second-vendor chain (GM, GM2, GM3): `arm_gm_preregistration.txt` at commit `37b3adb`, `arm_gm2_preregistration.txt` at `3019aab`, `arm_gm3_preregistration.txt` at `c0b5b97`. The gemma arm's (GM3) serving-path deviation, `arm_gm3_amendment1_openrouter.md` at `67cc4a1`, was registered before the affected cells were sampled. Cross-vendor arm (V): `arm_v_preregistration.md` at commit `5444f10`, ancestry-checked by its runners before any sampling; three amendments, each committed before the sampling it governs. First weak math-only arm (CC): `arm_cc_preregistration.txt` with frozen prompts, executor and smoke test, committed and pushed to the public remote at `285deca` before its first invocation. Replication arm (CC2) and Sonnet math-only arm (CCS): registered together in `arm_cc2_preregistration.txt` at commit `4d7b7c5`, pushed before sampling. That registration was written after the first weak math-only arm's results were known, and says so. Held-out-cells arm

(CN): `arm_cn_preregistration.txt` at commit `69a1642`, pushed before sampling, registered after the rest of the manuscript was complete. Its motivation is post hoc with respect to the paper's results, the registration says so, and its predictions were fixed before its first invocation. Container arm (CP) and recall arm (RP): `arm_cp_preregistration.txt` and `arm_rp_preregistration.txt`, each with its prompt-hash JSON and analysis script, registered together at commit `105e8b7`, pushed before the first invocation. The recall arm's one cell change before registration, $N = 32$ to $N = 30$, and its positive-control amendment at `15555eb` are documented in its registration. Paraphrase arm (PP): `arm_pp_preregistration.txt`, prompt hashes and scorer committed at `2e89614`, pushed before the first invocation; it discloses that its paraphrases were authored results-aware. The rectangle arm's (G) transfer was registered before collection and never refitted. Pinned-path arm (P) and library arm (CL): registrations, builders, frozen prompts, runners and analysis scripts, listed below, committed and pushed together at `aa473ad` before either was sampled. On multiplicity: twenty-six registered arms, and no correction across them. Each arm's predictions were registered with its own bars and falsifiers before its sampling, with the arm-S exception the deviations table records, and every arm is reported regardless of outcome. Three second-vendor instruments failed or stalled before one succeeded, the falsifier of Supplement S4 fired, and no arm was selected for inclusion on its result. The protection against the garden of forking paths is registration and exhaustive reporting, not a family-wise error rate. A reader who wants one can count the twenty-six arms in Table 4. That defence covers the registered predictions, not the post-hoc diagnostics: which unregistered analyses we ran is a degree of freedom registration does not constrain. Table 5 lists them. Three cut against us: the out-of-family sweep of Supplement S4, the rival-structure check of §5, and the wave stratification of Supplement S6.4.

- **Four limits on that claim.** (i) The digests cover the task prompt only. The subagent inherits instruction files outside it, so what is locked is a *fragment* of the conditioning context. A replicator cannot reconstruct those files, and we cannot establish that they were identical across the bare-square and trace windows, the boundary Supplement S6.4 identifies as a confound. (ii) `arm_f_prompts.json` covers $N = 13$, 17, 31, 35 and 37; the $N = 21$ bare hash is registered in `arm_t_preregistration.txt`. The $N = 43$ **bare hash appeared in no preregistration and no prompts file** until the pinned-path arm's, existing only in the ledger. It recomputes exactly from the bare template, but recomputation after the fact is not registration. (iii) **No hash was ever registered for the pilot prompt**, whose drift the registration describes only in prose; the ten pilot rows carry `prompt_sha256: null` in the corrected ledger with an explanatory note. (iv) The rectangle ledger, `arm_g_candidates.jsonl`, carries no prompt hashes, no run dates and no proposer fields.
- **Ledger metadata correction.** The shipped ledger's provenance fields contradicted the paper in three places. Trace rows carried the *bare* prompt hash for their cell; `run_date` was uniformly `2026-07-30` on all 215 rows; the Sonnet and `opus_alias` rows were stamped with the Haiku alias and dated id. All three are corrected in `arm_f_candidates_v2.jsonl`, metadata only. Every scored quantity, geometry, raw output, arm label and sample id is byte-identical to v1, verified field by field. Each corrected field carries a sibling `*_v1` field. `arm_f_candidates.jsonl` is unmodified, sha256 `02ecabcd...`; v2 is `9f845f78...`. Correction rules, sources, wave-boundary arithmetic and caveats, including that `run_date` has day granularity only, are in `ledger_v2_corrections.md`. Analysis scripts read v2.
- **Excluded and unreleased records.** The five concurrency-cap rejections appear in the ledger in no form, no `excluded_reason` field exists, and the exclusion rule was not registered; both rates are reported in Supplement S9.3. The `opus_alias` latency and token-count anomalies of §4.2 come from the runtime session transcript; no latency or token field exists in any released artifact.
- **Analysis artifacts.** Raw outputs are stored verbatim and scoring is deterministic and local. Bare square: `arm_f_raw.json`, `arm_f_candidates.jsonl`, `arm_f_candidates_v2.jsonl`. Rectangle: `arm_g_candidates.jsonl`. Extend cell: `arm_m_collect.jsonl`, scored by `arm_m_analysis.py`. Conditioning: `arm_mu_collect.jsonl`, scored by `arm_mu_analysis.py`, frozen output `arm_mu_results.txt`. Gemma: `arm_gm_gm3_checkpoint.jsonl`, every row's raw API response verbatim with serving path and provider tagged, scored by `arm_gm3_analysis.py`

into `arm_gm3_report.json` with both serving-path accountings. Value claims are gated on an independent LP oracle, not on the closed form being tested. The oracle, `n_sweep_forecast.py` with `verify_against_lp`, checks 83 configurations to within 10^{-9} and produced the negative result of Supplement S5.2. For the trace arm, Supplement S6, the faithfulness rubric `p7_faithfulness_rubric.md` was frozen by git commit `e181d2a` on 2026-08-01 at 03:56:56, before any rescoring. All 60 blind labels with justifications are in `p7_blind_labels.json`. Also included: `arm_t_analysis.py`, `n_sweep_forecast.json` and `rect_forecast.json` for the trace and rectangle arms, Supplement S6 and S5; `p11_mode_baseline.json`; `STATE.md`. Every table regenerates from raw outputs by a command documented in `HOW_TO_RUN.md`.

- **Internal checks.** Before submission the manuscript passed an internal adversarial check. Language models of other families, `deepseek-v4-pro`, `deepseek-v4-flash` and Gemini, under written protocols, independently recomputed the paper's statistics from the released ledger. Thirty-two claims were corrected as a result, itemized in the supplementary `corrections_ledger.md`. This is quality assurance, not peer review, and carries no external authority.

E.1 Per-arm ledgers and scripts

One entry per arm or registered group: ledgers, scripts, reports and the running corpus count. Letters stay in the headings so an auditor can join each entry to its row in the arm table of Appendix B.

- **The conditioning arm and the argmax-in-hand arm (MU, CH).** Raw rows verbatim in `arm_mu_collect.jsonl`: 180 of 180 launched invocations collected across the two arms, no runtime deaths. Scoring `arm_mu_analysis.py`, committed before sampling completed; frozen output `arm_mu_results.txt`. These arms bring the study corpus to 468 invocations.
- **The first weak math-only arm (CC).** Raw rows verbatim in `arm_cc_collect.jsonl`: 45 of 45 launched invocations collected, no runtime deaths. Scoring `arm_cc_analysis.py`; frozen report `arm_cc_report.json`; summary `arm_cc_results.txt`.
- **The pinned-path arm (P) and the thinking-budget arm (P-D).** Pinned-path arm: registration `arm_p_preregistration.txt`; prompts and hashes `arm_p_prompts.json`; runner `arm_p_run.py`; scorer `arm_p_analysis.py`; report `arm_p_report.json`. Ledger `arm_p_collect.jsonl`: 225 rows, six refused by the serving path. Post-hoc diagnostics `arm_p_diag_temperature.jsonl` and `arm_p_diag_runtime.jsonl`, labelled post hoc in their headers. Thinking-budget arm: registration `arm_pd_preregistration.txt` at commit `5554cc0`; builder `arm_pd_build.py`, the pinned-path arm's $N = 13$ prompt with hash asserted; spec `arm_pd_prompts.json`; runner `arm_pd_run.py`. Scorer `arm_pd_analysis.py`, the pinned-path scorer imported unmodified. Ledger `arm_pd_collect.jsonl`: 27 rows plus the 402 rows kept for resume. Report `arm_pd_report.json`, marked INTERIM.
- **The library arm (CL).** Registration `arm_cl_preregistration.txt`; builder `arm_cl_build.py`, one clause replaced in the weak math-only arms' prompt, hash asserted; prompts `arm_cl_prompts.json`; runner `arm_cl_run.py`. Scorer `arm_cl_analysis.py`, with the driver and the `numpy` and `scipy` versions recorded in the report. Ledger `arm_cl_collect.jsonl`: 90 rows. Report `arm_cl_report.json`. The post-hoc re-execution and lenient reparse are `recount_cl.py` and `cl_lenient_reparse.py` in the paper repository's `loop/` directory, outputs `cl_recount.json` and `cl_lenient.json`.
- **The isolating arm (CCP).** Registration `arm_ccp_preregistration.txt` at commit `ebf213e`; builder `arm_ccp_build.py`, the weak math-only arms' prompts byte-identical, hashes asserted; prompts `arm_ccp_prompts.json`. Runner `arm_ccp_run.py`, with the one-worker amendment after the row-24 interruption at `32fd3c7`. Scorer `arm_ccp_analysis.py`, the library arm's pipeline with the allowlist set to `math`. Ledger `arm_ccp_collect.jsonl`: 45 rows plus the one 402 row, both segments timestamped. Report `arm_ccp_report.json`.

- **The weak top-up (CL-W).** Registration arm_clw_preregistration.txt at commit f73222a; builder arm_clw_build.py, the weak library arm's prompts byte-identical, hashes asserted; prompts arm_clw_prompts.json; runner arm_clw_run.py. Scorer arm_clw_analysis.py, the library arm's pipeline over the pooled 90 rows, both readings per row. Ledger arm_clw_collect.jsonl: 45 rows. Report arm_clw_report.json, every row's registered and lenient result.
- **The optimizer-alone baseline (B).** Registration arm_b_preregistration.txt at commit 1932cc0; amendment arm_b_amendment1_fixed_restarts.md at c486843; program arm_b_baseline.py, sha asserted by the runner. Runner arm_b_run.py, the library arm's pipeline with no model. Ledger arm_b_collect.jsonl: 45 substituted sources. Report arm_b_report.json. The run the allowlist blocked is kept as arm_b_v1_blocked_collect.jsonl with its report and log. No model was invoked; the study corpus stays at 1500.
- **The replication arm (CC2) and the Sonnet math-only arm (CCS).** Ledgers arm_cc2_collect.jsonl and arm_ccs_collect.jsonl, 45 of 45 each, no runtime deaths. Runner arm_cc2_analysis.py, the first weak math-only arm's registered pipeline unmodified. These arms bring the study corpus to 603 invocations.